INCH-POUND

MIL-M-28787/535
20 SEPTEMBER 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

IOC DATA BUS UPPER MODULE, KEY CODE EWC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Input/Output Controller (IOC) data bus upper module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.364 failures per million operating hours for class I or 3.249 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be EWC.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vco stress	IV	S	---------Module shall pass all remaining tests-------						
D1	Ibd	V	S	0.0	7.0	0.0	7.0	0.0	7.0	mA
D2	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D3-D5	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D6	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D7	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D8, D9	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D10	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D11	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	mA
D12	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D13	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
D14	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D15	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D16-D25	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D26, D27	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D28	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D29	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D30	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D31, D32	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D33, D34	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D35	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D36	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	mA
E1	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E2	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E3, E4	Ios	VI	S	-225 0	-50.0	-225.0	-50 0	-225.0	-50 0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E5-E20	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E21	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E22-E29	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E30-E45	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E46-E49	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E50	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E51	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E52	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E53	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E54	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E55	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E56	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E57-E61	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E62	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E63	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E64	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E65	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E66	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E67	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E68	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E69	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E70-E77	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
F1	Iih	VII	X	-14.0	0.35	-14.0	0.35	-14.0	0.35	mA
F2	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F3-F5	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F6	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F7	Iih	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F8,F9	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F10	Iih	VII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA
F11	Iih	VII	X	-2.6	0.1	-2.6	0.1	-2.6	0.1	mA
F12	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F13	Iih	VII	X	-12.0	0.3	-12.0	0.3	-12.0	0.3	mA
F14	Iih	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F15	Ith	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F16-F19	Ith	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F20-F23	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F24-F27	Ith	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F28-F32	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F33	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F34,F35	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F36	Ith	VII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
F37	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F38	Ith	VII	X	-2.6	0.06	-2.6	0.06	-2.6	0.06	mA
F39-F42	Ith	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F43-F46	Ith	VII	X	-0.87	0.11	-0.87	0.11	-0.87	0.11	mA
F47	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F48	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F49,F50	Ith	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F51,F52	Ith	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F53	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F54,F55	Ith	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F56-F59	Ith	VII	X	-0.87	0.11	-0.87	0.11	-0.87	0.11	mA
F60	Ith	VII	X	-5.2	0.14	-5.2	0.14	-5.2	0.14	mA
F61-F68	Ith	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G1	Iil	VIII	X	-14.0	0.35	-14.0	0.35	-14.0	0.35	mA
G2	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G3-G5	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G6	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G7	Iil	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G8,G9	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G10	Iil	VIII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA
G11	Iil	VIII	X	-2.6	0.1	-2.6	0.1	-2.6	0.1	mA
G12	Iil	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G13	Iil	VIII	X	-12.0	0.3	-12.0	0.3	-12.0	0.3	mA
G14	Iil	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G15	Iil	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 + 5°C 60 ∓ 5°C		Class II 4/ -55 + 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
G16-G19	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G20-G23	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G24-G27	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G28-G32	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G33	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G34,G35	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G36	Iil	VIII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
G37	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G38	Iil	VIII	X	-2.6	0.06	-2.6	0.06	-2.6	0.06	mA
G39-G42	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G43-G46	Iil	VIII	X	-0.87	0.11	-0.87	0.11	-0.87	0.11	mA
G47	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G48	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G49,G50	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G51,G52	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G53	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G54,G55	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G56-G59	Iil	VIII	X	-0.87	0.11	-0.87	0.11	-0.87	0.11	mA
G60	Iil	VIII	X	-5.2	0.14	-5.2	0.14	-5.2	0.14	mA
G61-G68	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
H1-H77	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J75	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K16	Iozh	XI	X	-0.04	0.04	-0.04	0.04	-0.04	0.04	mA
K17-K32	Iozl	XI	X	-0.04	0.04	-0.04	0.04	-0.04	0.04	mA
L1-L11	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L12	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L13,L14	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L15	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L16-L24	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L25	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
L26-L28	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L29	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L30,L31	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L32	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L33-L36	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	5.6	10.4	5.6	10.4	5.6	10.4	uF
M2	PSFC	XIII	S	6.3	11.7	6.3	11.7	6.3	11.7	uF
N	Icc	XIV	X	400	960	400	960	400	960	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	Ci/o	XVIII	Q	10	150	10	150	10	150	pF
R2	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R3	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R4,R5	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R6	Ci/o	XVIII	Q	6	75	6	75	6	75	pF
R7	Ci/o	XVIII	Q	8	100	8	100	8	100	pF
R8,R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10	Ci/o	XVIII	Q	6	150	6	150	6	150	pF
R11	Ci/o	XVIII	Q	10	150	10	150	10	150	pF
R12	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R13	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R14	Ci/o	XVIII	Q	6	150	6	150	6	150	pF
R15	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R16	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R17	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R18-R23	Ci/o	XVIII	Q	2	75	2	75	2	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R24,R25	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R26	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R27	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R28	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R29	C1/o	XVIII	Q	2	125	2	125	2	125	pF
R30	C1/o	XVIII	Q	6	100	8	100	6	100	pF
R31,R32	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R33,R34	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R35	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R36	C1/o	XVIII	Q	2	150	2	150	2	150	pF
R37-R44	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R45-R59	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R60	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R61	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R62-R64	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R65,R66	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R67	C1/o	XVIII	Q	8	125	8	125	8	125	pF
R68	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R69-R72	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R73-R76	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R77-R84	C1/o	XVIII	Q	4	75	4	75	4	75	pF
T1,T2	PSTA	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/535-1 for class I and M28787/535-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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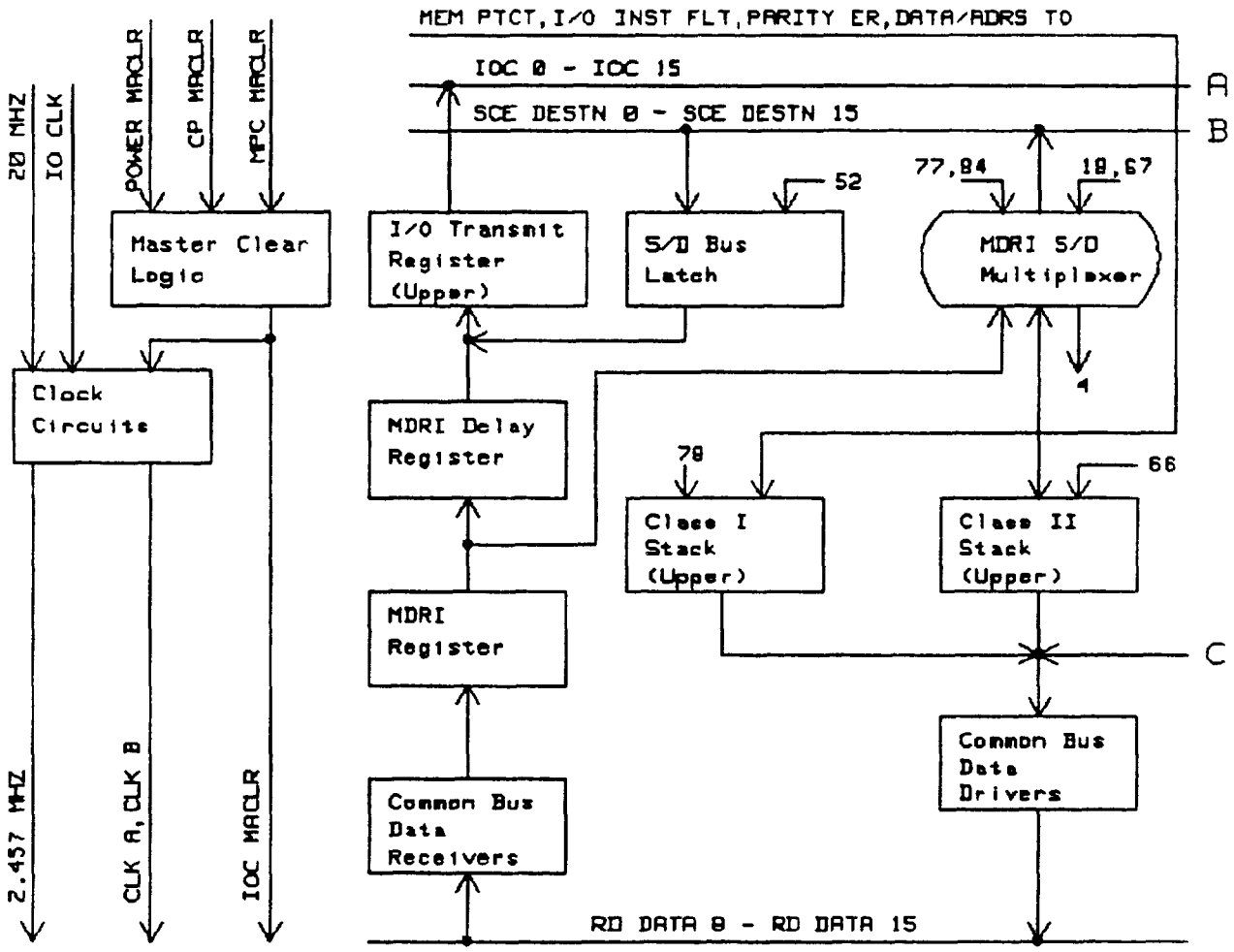


FIGURE 1. EWC functional block diagram.

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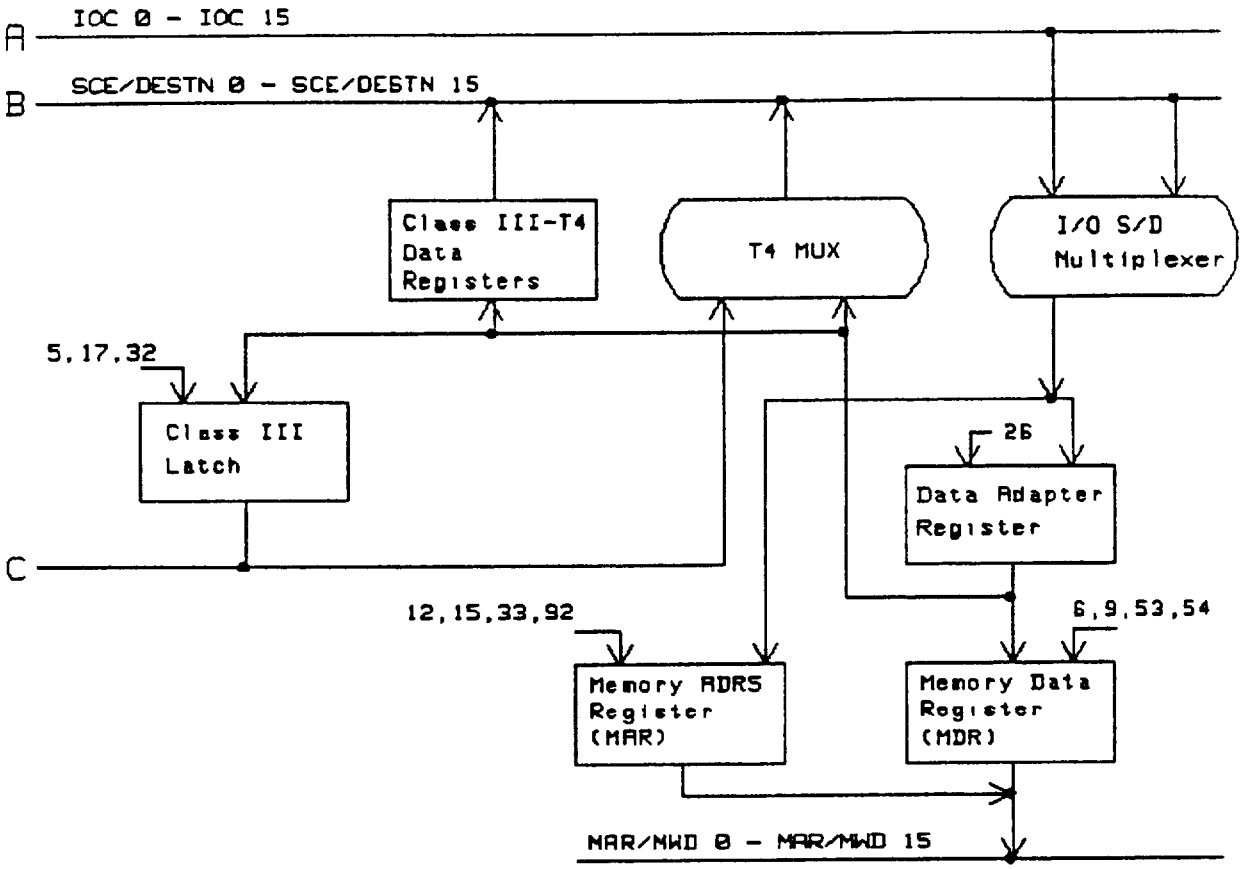


FIGURE 1. EWC functional block diagram - Continued.

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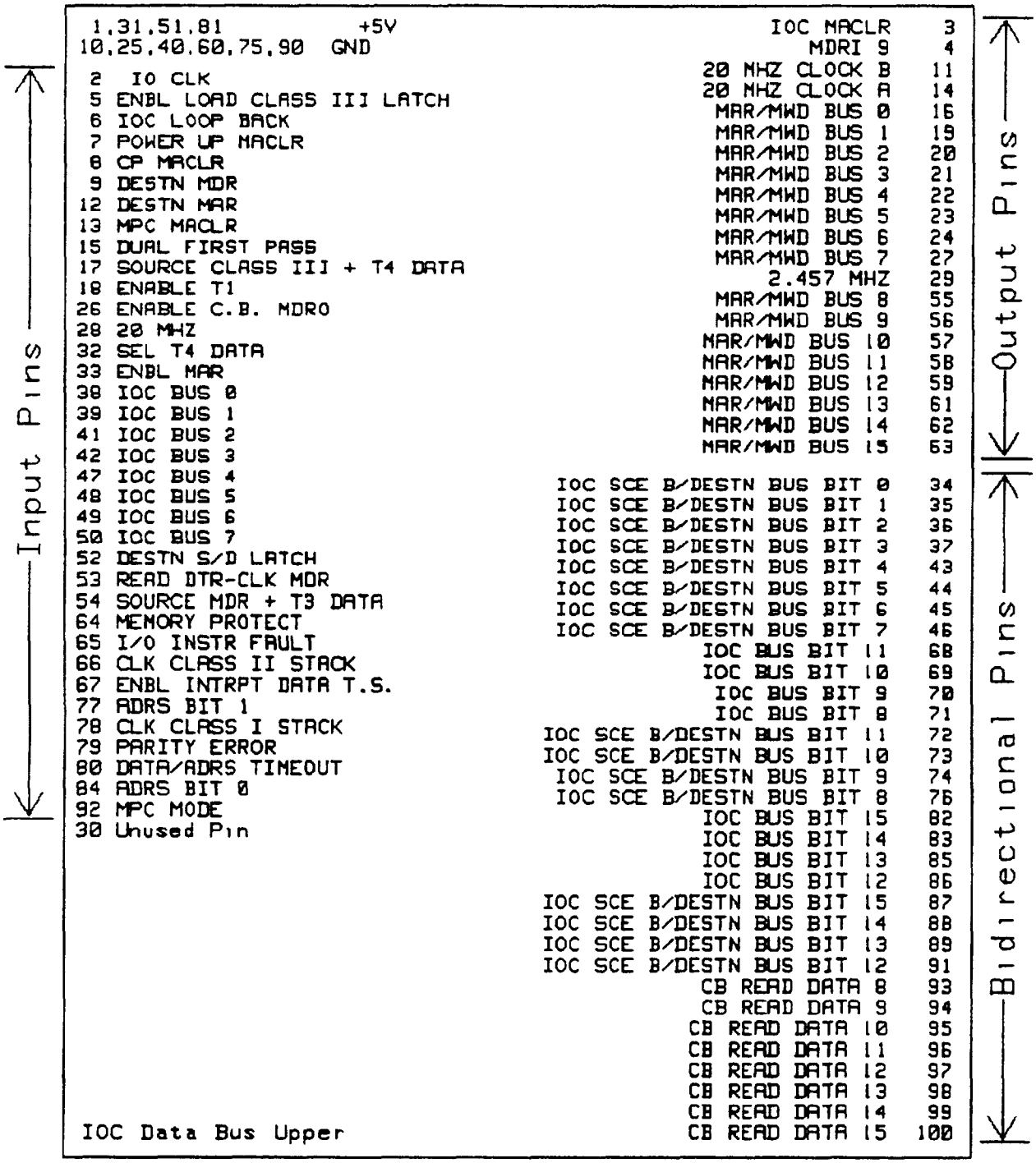


FIGURE 2. EWC pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

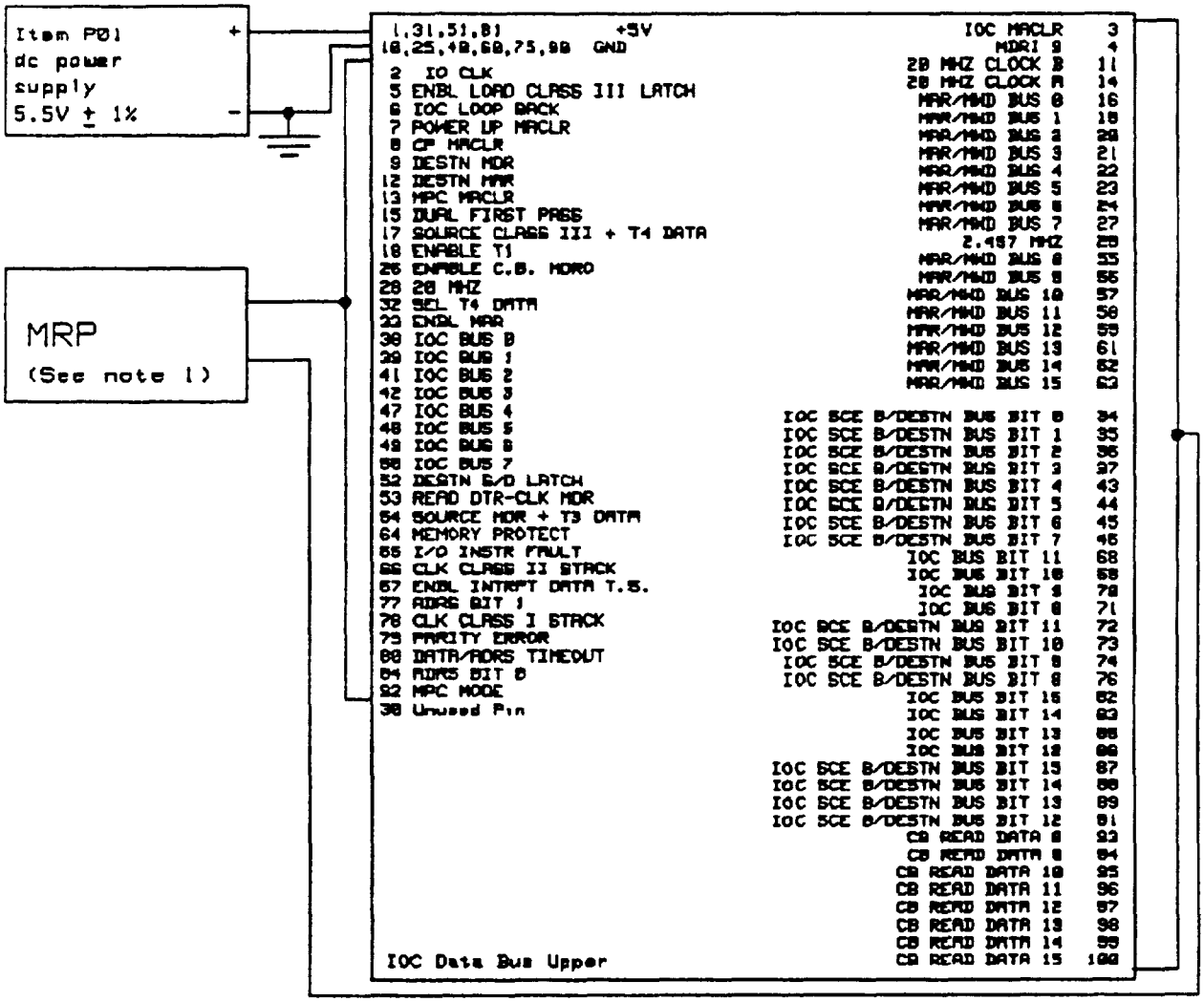
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. EWC operating temperature and component temperature test circuit.

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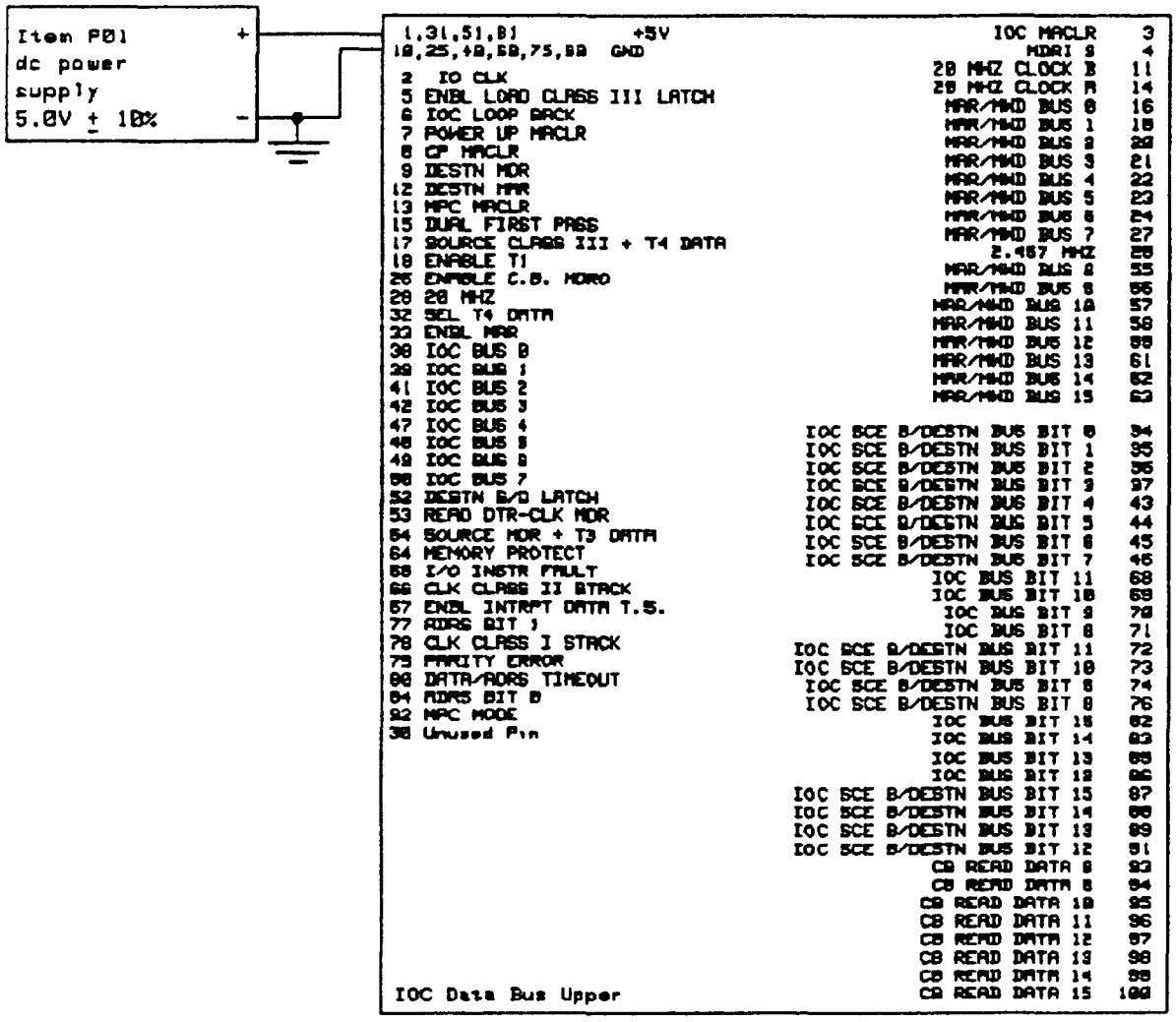


FIGURE 4. EWC vibration test circuit.

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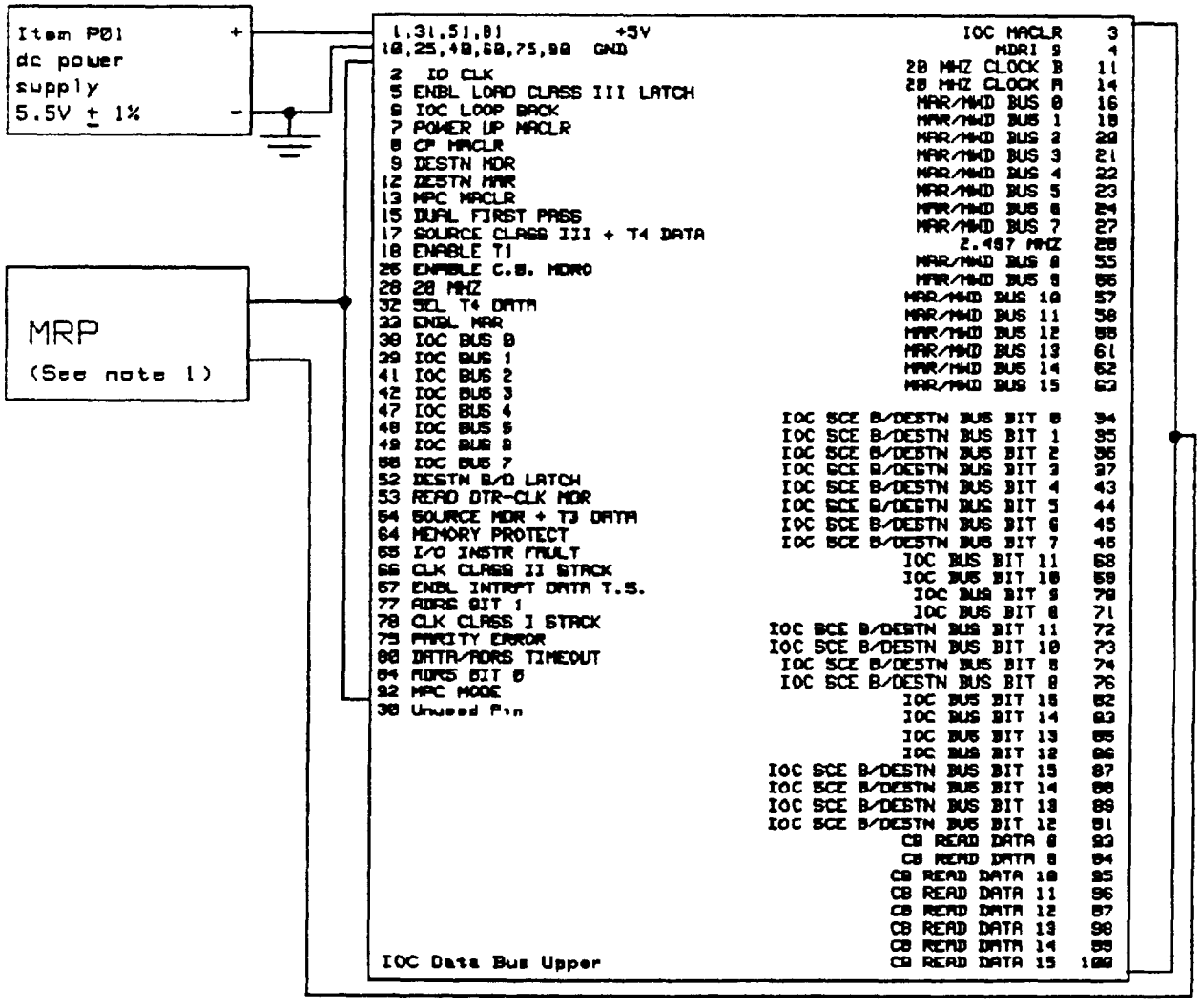
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. EWC life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 5.28 W (max); 3.54 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IO CLK	-1.2	0.4	2.4	-14.0	0.35	7.0	150
ENBL CIII	-1.2	0.4	2.4	-2.0	0.05	1.0	75
IOC LP BK	-1.2	0.4	2.4	-0.6	0.02	0.1	50
POWER MACLR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CP MACLR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DESTN MDR	-1.2	0.4	2.4	-1.8	0.06	0.3	75
DESTN MAR	-1.2	0.4	2.4	-2.4	0.08	0.4	100
MPC MACLR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DUAL 1ST	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CLIII + T4	-1.2	0.4	2.4	-1.4	0.06	0.3	150
ENBL T1	-1.2	0.4	2.4	-2.6	0.1	0.5	150
ENBL MDRO	-1.5	0.4	2.4	-0.4	0.04	0.2	100
20 MHZ	-1.2	0.4	2.4	-12.0	0.3	3.0	100
SEL T4	-1.2	0.4	2.4	-2.4	0.08	0.4	150

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ENBL MAR	-1.5	0.4	2.4	-0.4	0.04	0.2	100
SCE/DESTN 0	NA	0.4	2.4	-0.65	0.07	NA	75
SCE/DESTN 1	NA	0.4	2.4	-0.65	0.07	NA	75
SCE/DESTN 2	NA	0.4	2.4	-0.65	0.07	NA	75
SCE/DESTN 3	NA	0.4	2.4	-0.65	0.07	NA	75
IOC 0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOC 1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IOC 2	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOC 3	-1.2	0.4	2.4	-0.6	0.02	0.1	75
SCE/DESTN 4	NA	0.4	2.4	-0.65	0.07	NA	75
SCE/DESTN 5	NA	0.4	2.4	-0.65	0.07	NA	75
SCE/DESTN 6	NA	0.4	2.4	-0.65	0.07	NA	75
SCE/DESTN 7	NA	0.4	2.4	-0.65	0.07	NA	75
IOC 4	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC 5	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOC 6	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOC 7	-1.2	0.4	2.4	-0.6	0.02	0.1	75
S/D LTCH	-1.2	0.4	2.4	-0.6	0.02	0.1	100
RD DTR-CLK	-1.5	0.4	2.4	-0.2	0.02	0.1	100
MDR + T3	-1.2	0.4	2.4	-1.2	0.04	0.2	75
MEM PTCT	-1.2	0.4	2.4	-1.2	0.04	0.2	100
I/O INST FLT	-1.2	0.4	2.4	-3.2	0.09	1.2	125
CLK CII	-1.5	0.4	2.4	-0.2	0.02	0.1	125
ENBL INTR	-1.2	0.4	2.4	-2.6	0.06	0.3	100
IOC 11	NA	0.4	2.4	-0.65	0.07	NA	75
IOC 10	NA	0.4	2.4	-0.65	0.07	NA	100
IOC 9	NA	0.4	2.4	-0.65	0.07	NA	100
IOC 8	NA	0.4	2.4	-0.65	0.07	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SCE/DESTN 11	NA	0.4	2.4	-0.87	0.11	NA	150
SCE/DESTN 10	NA	0.4	2.4	-0.87	0.11	NA	125
SCE/DESTN 9	NA	0.4	2.4	-0.87	0.11	NA	125
SCE/DESTN 8	NA	0.4	2.4	-0.87	0.11	NA	150
ADRS 1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CLK CI	-1.5	0.4	2.4	-0.2	0.02	0.1	75
PARITY ER	-1.2	0.4	2.4	-0.8	0.04	0.2	75
DATA/ADRS TO	-1.2	0.4	2.4	-0.8	0.04	0.2	75
IOC 15	NA	0.4	2.4	-0.65	0.07	NA	75
IOC 14	NA	0.4	2.4	-0.65	0.07	NA	75
ADRS 0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOC 13	NA	0.4	2.4	-0.65	0.07	NA	75
IOC 12	NA	0.4	2.4	-0.65	0.07	NA	75
SCE/DESTN 15	NA	0.4	2.4	-0.87	0.11	NA	150

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SCE/DESTN 14	NA	0.4	2.4	-0.87	0.11	NA	150
SCE/DESTN 13	NA	0.4	2.4	-0.87	0.11	NA	150
SCE/DESTN 12	NA	0.4	2.4	-0.87	0.11	NA	150
MPC MODE	-1.2	0.4	2.4	-5.2	0.14	2.2	150
RD DATA 8	NA	0.4	2.4	-0.65	0.07	NA	75
RD DATA 9	NA	0.4	2.4	-0.65	0.07	NA	75
RD DATA 10	NA	0.4	2.4	-0.65	0.07	NA	75
RD DATA 11	NA	0.4	2.4	-0.65	0.07	NA	75
RD DATA 12	NA	0.4	2.4	-0.65	0.07	NA	75
RD DATA 13	NA	0.4	2.4	-0.65	0.07	NA	75
RD DATA 14	NA	0.4	2.4	-0.65	0.07	NA	75
RD DATA 15	NA	0.4	2.4	-0.65	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOC MACLR	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
MDRI 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CLK B	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
CLK A	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
MAR/MWD 0	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
MAR/MWD 1	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
MAR/MWD 2	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
MAR/MWD 3	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
MAR/MWD 4	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
MAR/MWD 5	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
MAR/MWD 6	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
MAR/MWD 7	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
2.457 MHZ	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
SCE/DESTN 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SCE/DESTN 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
SCE/DESTN 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
SCE/DESTN 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
SCE/DESTN 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
SCE/DESTN 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
SCE/DESTN 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
SCE/DESTN 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MAR/MWD 8	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	75
MAR/MWD 9	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	75
MAR/MWD 10	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	75
MAR/MWD 11	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	75
MAR/MWD 12	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	75
MAR/MWD 13	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	75
MAR/MWD 14	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MAR/MWD 15	2.4	0.5	-0.16	1.6	0.04	0.04	-30.0	100
IOC 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
SCE/DESTN 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	150
SCE/DESTN 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	125
SCE/DESTN 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	125
SCE/DESTN 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	150
IOC 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SCE/DESTN 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	150
SCE/DESTN 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	150
SCE/DESTN 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	150
SCE/DESTN 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0 -60.0	150
RD DATA 8	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
RD DATA 9	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
RD DATA 10	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
RD DATA 11	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
RD DATA 12	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
RD DATA 13	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
RD DATA 14	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
RD DATA 15	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CLIII + T4 to SCE/DESTN 0	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 1	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 2	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 3	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 4	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 5	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 6	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 7	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 8	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 9	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 10	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 11	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 12	-	-	-	32	32	32	nS	12a

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CLIII + T4 to SCE/DESTN 13	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 14	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 15	-	-	-	32	32	32	nS	12a
2	tpd	IO CLK to MAR/MWD 0	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 1	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 2	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 3	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 4	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 5	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 6	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 7	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 8	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 9	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 10	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 11	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 12	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 13	-	-	-	36	36	36	nS	12a

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	IO CLK to MAR/MWD 14	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 15	-	-	-	36	36	36	nS	12a
3	tpd	RD DTR-CLK to MDRI 9	-	-	-	26	26	26	nS	12a
4	tpd	IO CLK to IOC 8	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 9	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 10	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 11	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 12	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 13	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 14	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 15	-	-	-	28	28	28	nS	12a
5	tpd	IO CLK to RD DATA 8	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 9	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 10	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 11	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 12	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 13	-	-	-	51	51	51	nS	12a
6	tsu	ENBL T1 to IO CLK	12	12	12	-	-	-	nS	12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	th	IO CLK to ENBL T1	14	14	14	-	-	-	nS	12b
8	tsu	IOC 0 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 1 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 2 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 3 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 4 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 5 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 6 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 7 to IO CLK	24	24	24	-	-	-	nS	12b
9	th	IO CLK to IOC 0	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 1	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 2	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 3	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 4	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 5	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 6	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 7	11	11	11	-	-	-	nS	12b

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6.4 Module function. The IOC data bus interface is contained on two modules, IOC data bus upper (EWC) and IOC data bus lower (AWG). The IOC data bus upper and the IOC data bus lower modules function to interface the common bus to the IOC bus. The IOC data bus interface modules contain transmit registers, Memory Data Register Input (MDRI), the class I, II, and III interrupt stacks, Memory Address Register (MAR), Memory Data Register (MDR), error capture circuits, and interrupt request flip-flops. The transmit registers are used to transmit data and commands to the IOCs via the IOC bus. Data or commands that are destined for the Input/Output Channel Adapters (IOCAs) will first be placed into the MDRI registers. The class I, II, and III interrupt stacks hold interrupt data until sampled by the Central Processor (CP) at its convenience. The class I stack holds parity error and data/address timeout interrupts; the class II stack holds Input/Output (I/O) instruction fault and protect interrupt data; and the class III stack holds channel data along with Intercomputer Timeout (ICTO), External Interrupt (EI), and input or output chain interrupts. IOC data bus upper and lower modules contain bits 15-8 and bits 7-0 of the interrupt word, respectively. The 16-bit MAR and MDR, both contained on the IOC data bus upper module, are used to hold the address and data destined for main memory during Externally Specified Addressing (ESA) mode and for writes. These registers will only be used during the buffer mode. Error capture circuits receive abort, timeout, and parity error information which is used to clock the class I, II, and III interrupt flip-flops. Error information is also placed onto the IOC bus for transfer to the IOCAs. The interrupt flip-flop outputs are used to send interrupt requests to the CP emulation control 2 module. IOC data bus upper (EWC) contains common bus data drivers, interrupt stacks and control circuits, MDR, MAR, and associated logic and control circuits.

6.4.1 Common bus data drivers. The common bus data drivers receive wire-ORed interrupt information from the class I, II, and III interrupt stacks and send the data to the CP. The drivers are enabled by the enable interrupt data signal from the IOC common bus control module.

6.4.2 Interrupt stacks. The class I and class II interrupt stacks receive their inputs from the IOC data bus lower module. Control for the interrupt stacks is provided by common bus address bits 0 and 1. The stacks are clocked by the interrupt request control circuits (IOC data bus lower module). Outputs from the class I and II interrupt stacks are sent to the common bus via the common bus data drivers. The class III interrupt stack receives adapter data bits 5-0 as inputs. The

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buffer circuit for class III data is always enabled and control is exercised by common bus address bits 0 and 1. The class III stack outputs are sent to the common bus via the common bus data drivers.

6.4.3 Source B/destination multiplexer. The source B/destination multiplexer selects either common bus data (bits 15-8) or interrupt data and destination/mode information to place on the source B/destination bus. Selection is done by microinstruction bit 8. The source B/destination multiplexer is enabled by the source MDR + T3 data signal from the IOC sequencer module when the microinstruction Source B (SB) field = 14.

6.4.4 Source/Destination IOC multiplexer. The constantly enabled source destination IOC multiplexer provides the capability to select either IOC bus bit 15-0 or source B/destination bus bits 15-0 for application to the data adapter latch or the MAR. Selection is controlled by the IOC sequencer module via the register control circuits.

6.4.5 T4 multiplexer. The T4 multiplexer receives interrupt information from the class III latch or adapter data. The multiplexer enable signal originates at the IOC sequencer from a microinstruction with SB = 13. Selection is accomplished by the sel T4 data signal from the IOC channel control module which uses microinstruction bits 6-4 for this purpose.

6.4.6 MDRI and MDRI delay registers. Common bus data bits 15-8 are clocked into the MDRI register by the read DTR-clock MDR signal which originates at the IOC common bus control module. The MDRI delay register is clocked by the IO clock and T3. Its outputs are transmitted to the IOC bus at T1 and T2 times.

6.4.7 Memory Address Register (MAR). MAR holds the memory address prior to the address being sent out on the MAR/memory write data bus. The MAR is enabled by MAR control circuits on the IOC common bus control module. The MAR is clocked by the gated IO clock and MAR destination bit from the IOC sequencer module.

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6.4.8 Description of I/O pins.

Signal name	Pin number	I/O	Description
IO CLK	2	I	IO CLK
IOC MACLR	3	O	IOC MACLR
MDRI 9	4	O	MDRI 9
ENBL CIII	5	I	ENBL LOAD CLASS III LATCH
IOC LP BK	6	I	IOC LOOP BACK
POWER MACLR	7	I	POWER UP MACLR
CP MACLR	8	I	CP MACLR
DESTN MDR	9	I	DESTN MDR
CLK B	11	O	20 MHZ CLOCK B
DESTN MAR	12	I	DESTN MAR
MPC MACLR	13	I	MPC MACLR
CLK A	14	O	20 MHZ CLOCK A
DUAL 1ST	15	I	DUAL FIRST PASS
MAR/MWD 0	16	O	MAR/MWD BUS 0
CLIII + T4	17	I	SOURCE CLASS III + T4 DATA
ENBL T1	18	I	ENABLE T1
MAR/MWD 1	19	O	MAR/MWD BUS 1
MAR/MWD 2	20	O	MAR/MWD BUS 2
MAR/MWD 3	21	O	MAR/MWD BUS 3
MAR/MWD 4	22	O	MAR/MWD BUS 4

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MAR/MWD 5	23	O	MAR/MWD BUS 5
MAR/MWD 6	24	O	MAR/MWD BUS 6
ENBL MDRO	26	I	ENBL C.B. MDRO
MAR/MWD 7	27	O	MAR/MWD BUS 7
20 MHZ	28	I	20 MHZ
2.457 MHZ	29	O	2.457 MHZ
SEL T4	32	I	SEL T4 DATA
ENBL MAR	33	I	ENBL MAR
SCE/DESTN 0	34	I/O	IOC SCE B/DESTN BUS BIT 0
SCE/DESTN 1	35	I/O	IOC SCE B/DESTN BUS BIT 1
SCE/DESTN 2	36	I/O	IOC SCE B/DESTN BUS BIT 2
SCE/DESTN 3	37	I/O	IOC SCE B/DESTN BUS BIT 3
IOC 0	38	I	IOC BUS 0
IOC 1	39	I	IOC BUS 1
IOC 2	41	I	IOC BUS 2
IOC 3	42	I	IOC BUS 3
SCE/DESTN 4	43	I/O	IOC SCE B/DESTN BUS BIT 4
SCE/DESTN 5	44	I/O	IOC SCE B/DESTN BUS BIT 5
SCE/DESTN 6	45	I/O	IOC SCE B/DESTN BUS BIT 6
SCE/DESTN 7	46	I/O	IOC SCE B/DESTN BUS BIT 7

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC 4	47	I	IOC BUS 4
IOC 5	48	I	IOC BUS 5
IOC 6	49	I	IOC BUS 6
IOC 7	50	I	IOC BUS 7
S/D LTCH	52	I	DESTN S/D LATCH
RD DTR-CLK	53	I	READ DTR-CLK MDR
MDR + T3	54	I	SOURCE MDR + T3 DATA
MAR/MWD 8	55	O	MAR/MWD BUS 8
MAR/MWD 9	56	O	MAR/MWD BUS 9
MAR/MWD 10	57	O	MAR/MWD BUS 10
MAR/MWD 11	58	O	MAR/MWD BUS 11
MAR/MWD 12	59	O	MAR/MWD BUS 12
MAR/MWD 13	61	O	MAR/MWD BUS 13
MAR/MWD 14	62	O	MAR/MWD BUS 14
MAR/MWD 15	63	O	MAR/MWD BUS 15
MEM PTCT	64	I	MEMORY PROTECT
I/O INST FLT	65	I	I/O INSTR FAULT
CLK CII	66	I	CLK CLASS II STACK
ENBL INTR	67	I	ENBL INTRPT DATA T.S.
IOC 11	68	I/O	IOC BUS BIT 11

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC 10	69	I/O	IOC BUS BIT 10
IOC 9	70	I/O	IOC BUS BIT 9
IOC 8	71	I/O	IOC BUS BIT 8
SCE/DESTN 11	72	I/O	IOC SCE B/DESTN BUS BIT 11
SCE/DESTN 10	73	I/O	IOC SCE B/DESTN BUS BIT 10
SCE/DESTN 9	74	I/O	IOC SCE B/DESTN BUS BIT 9
SCE/DESTN 8	76	I/O	IOC SCE B/DESTN BUS BIT 8
ADRS 1	77	I	ADRS BIT 1
CLK CI	78	I	CLK CLASS I STACK
PARITY ER	79	I	PARITY ERROR
DATA/ADRS TO	80	I	DATA/ADRS TIMEOUT
IOC 15	82	I/O	IOC BUS BIT 15
IOC 14	83	I/O	IOC BUS BIT 14
ADRS 0	84	I	ADRS BIT 0
IOC 13	85	I/O	IOC BUS BIT 13
IOC 12	86	I/O	IOC BUS BIT 12
SCE/DESTN 15	87	I/O	IOC SCE B/DESTN BUS BIT 15
SCE/DESTN 14	88	I/O	IOC SCE B/DESTN BUS BIT 14
SCE/DESTN 13	89	I/O	IOC SCE B/DESTN BUS BIT 13
SCE/DESTN 12	91	I/O	IOC SCE B/DESTN BUS BIT 12

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MPC MODE	92	I	MPC MODE
RD DATA 8	93	I/O	CB READ DATA 8
RD DATA 9	94	I/O	CB READ DATA 9
RD DATA 10	95	I/O	CB READ DATA 10
RD DATA 11	96	I/O	CB READ DATA 11
RD DATA 12	97	I/O	CB READ DATA 12
RD DATA 13	98	I/O	CB READ DATA 13
RD DATA 14	99	I/O	CB READ DATA 14
RD DATA 15	100	I/O	CB READ DATA 15

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N730)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,30	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	C	30	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 KOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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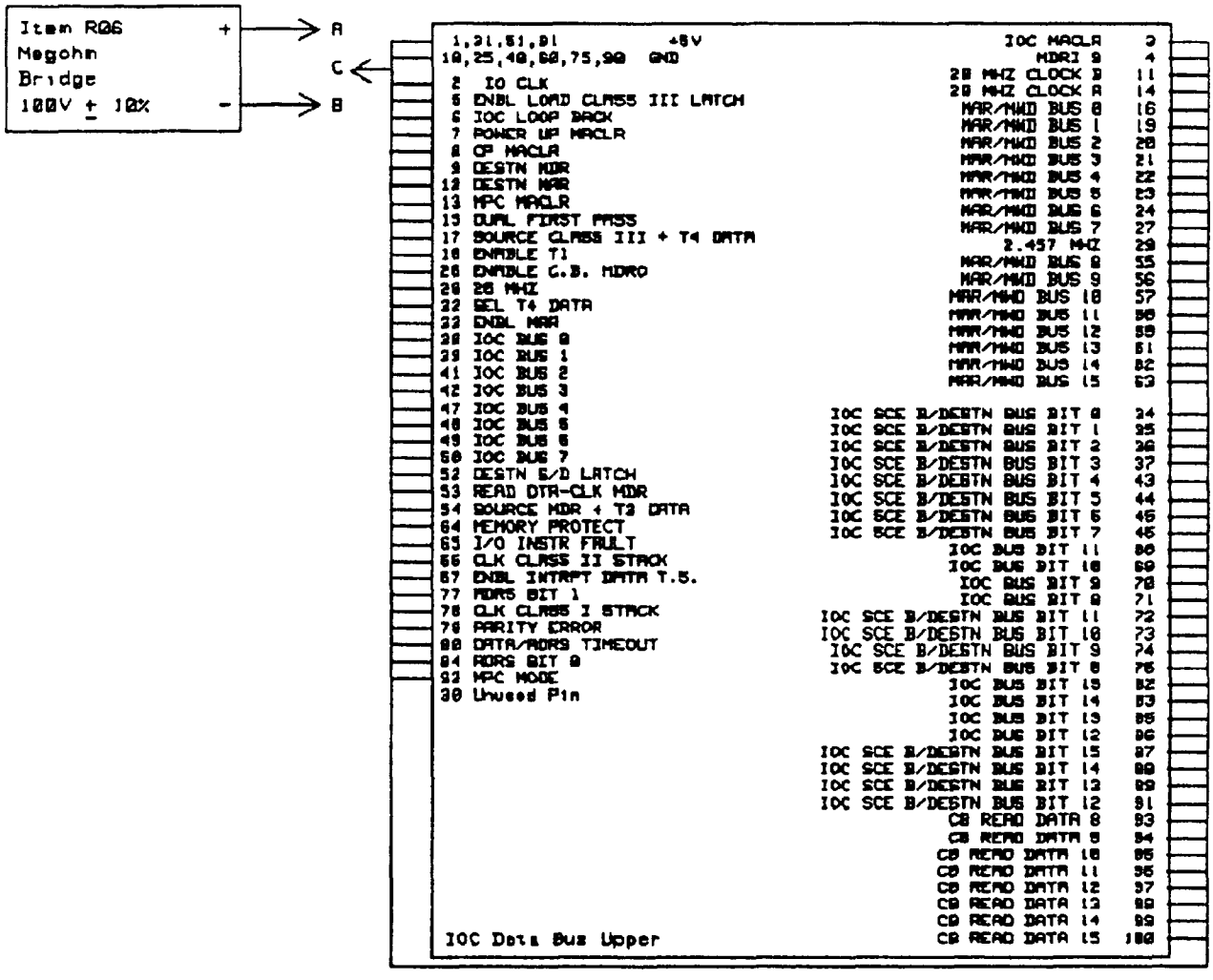


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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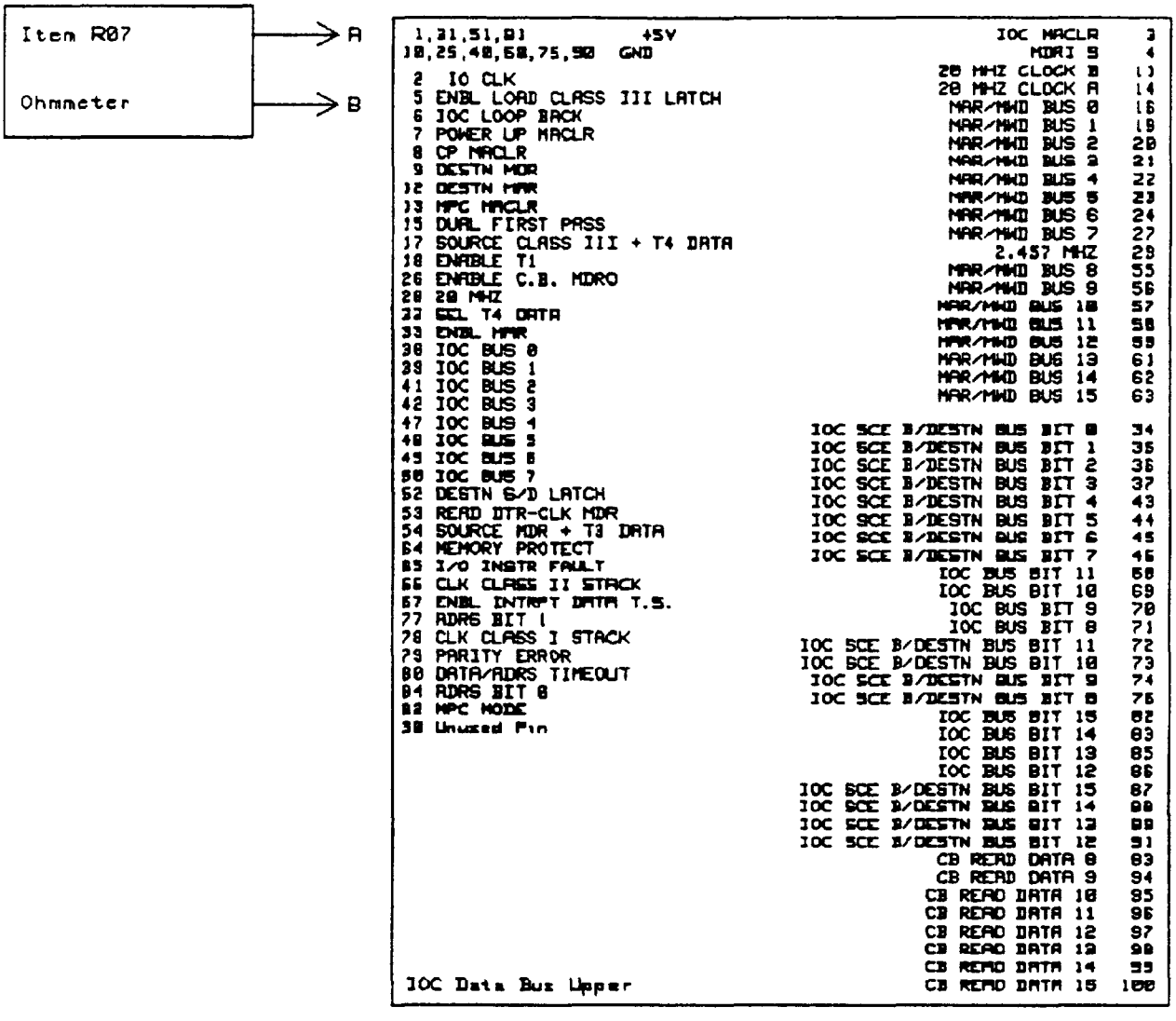


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.73 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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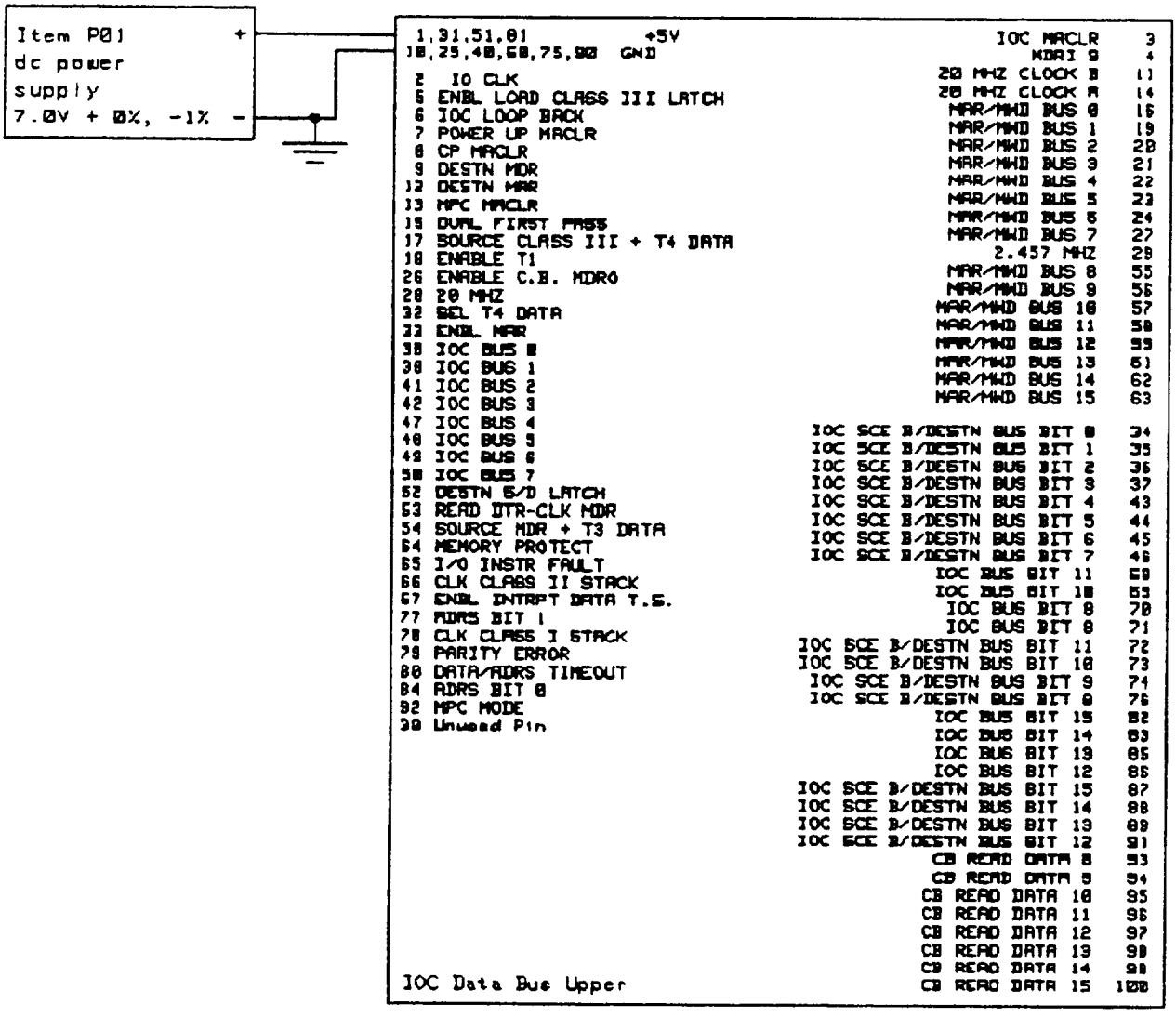


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	IO CLK	2	5.5	0.0	7.0	mA	1-3
D2	ENBL CIII	5	5.5	0.0	1.0	mA	1-3
D3	IOC LP BK	6	5.5	0.0	0.1	mA	1-3
D4	POWER MACLR	7					
D5	CP MACLR	8	5.5	0.0	0.1	mA	1-3
D6	DESTN MDR	9	5.5	0.0	0.3	mA	1-3
D7	DESTN MAR	12	5.5	0.0	0.4	mA	1-3
D8	MPC MACLR	13	5.5	0.0	0.1	mA	1-3
D9	DUAL 1ST	15	5.5	0.0	0.1	mA	1-3
D10	CLIII + T4	17	5.5	0.0	0.3	mA	1-3
D11	ENBL T1	18	5.5	0.0	0.5	mA	1-3
D12	ENBL MDRO	26	5.5	0.0	0.2	mA	1-3
D13	20 MHZ	28	5.5	0.0	3.0	mA	1-3
D14	SEL T4	32	5.5	0.0	0.4	mA	1-3
D15	ENBL MAR	33	5.5	0.0	0.2	mA	1-3
D16	IOC 0	38	5.5	0.0	0.1	mA	1-3
D17	IOC 1	39					
D18	IOC 2	41					
D19	IOC 3	42					
D20	IOC 4	47	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	IOC 5	48	5.5	0.0	0.1	mA	1-3
D22	IOC 6	49					
D23	IOC 7	50					
D24	S/D LTCH	52					
D25	RD DTR-CLK	53	5.5	0.0	0.1	mA	1-3
D26	MDR + T3	54	5.5	0.0	0.2	mA	1-3
D27	MEM PTCT	64	5.5	0.0	0.2	mA	1-3
D28	I/O INST FLT	65	5.5	0.0	1.2	mA	1-3
D29	CLK CII	66	5.5	0.0	0.1	mA	1-3
D30	ENBL INTR	67	5.5	0.0	0.3	mA	1-3
D31	ADRS 1	77	5.5	0.0	0.1	mA	1-3
D32	CLK CI	78	5.5	0.0	0.1	mA	1-3
D33	PARITY ER	79	5.5	0.0	0.2	mA	1-3
D34	DATA/ADRS TO	80	5.5	0.0	0.2	mA	1-3
D35	ADRS 0	84	5.5	0.0	0.1	mA	1-3
D36	MPC MODE	92	5.5	0.0	2.2	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D36
Power supply current limit	1.44 A	NA	D1-D36
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D36
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D36
low, (VL)	0.0 V	± 100 mV	D1-D36
Input current measurement	NA	± 350 uA	D1
	NA	± 50 uA	D2
	NA	± 5 uA	D3-D5
	NA	± 15 uA	D6
	NA	± 20 uA	D7
	NA	± 5 uA	D8,D9
	NA	± 15 uA	D10
	NA	± 25 uA	D11
	NA	± 10 uA	D12
	NA	± 150 uA	D13
	NA	± 20 uA	D14
	NA	± 10 uA	D15
	NA	± 5 uA	D16-D25
	NA	± 10 uA	D26,D27
	NA	± 60 uA	D28
	NA	± 5 uA	D29
	NA	± 15 uA	D30
	NA	± 5 uA	D31,D32
	NA	± 10 uA	D33,D34
	NA	± 5 uA	D35
	NA	± 110 uA	D36

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	IOC MACLR	3	0.0	-225.0	-50.0	mA	1-4
E2	MDRI 9	4	0.0	-260.0	-30.0	mA	1-4
E3	CLK B	11	0.0	-225.0	-50.0	mA	1-4
E4	CLK A	14	0.0	-225.0	-50.0	mA	1-4
E5	MAR/MWD 0	16	0.0	-260.0	-30.0	mA	1-4
E6	MAR/MWD 0	16					
E7	MAR/MWD 1	19					
E8	MAR/MWD 1	19					
E9	MAR/MWD 2	20					
E10	MAR/MWD 2	20					
E11	MAR/MWD 3	21					
E12	MAR/MWD 3	21					
E13	MAR/MWD 4	22					
E14	MAR/MWD 4	22					
E15	MAR/MWD 5	23					
E16	MAR/MWD 5	23					
E17	MAR/MWD 6	24					
E18	MAR/MWD 6	24					
E19	MAR/MWD 7	27					
E20	MAR/MWD 7	27	0.0	-260.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	2.457 MHZ	29	0.0	-225.0	-50.0	mA	1-4
E22	SCE/DESTN 0	34	0.0	-150.0	-60.0	mA	1-4
E23	SCE/DESTN 1	35					
E24	SCE/DESTN 2	36					
E25	SCE/DESTN 3	37					
E26	SCE/DESTN 4	43					
E27	SCE/DESTN 5	44					
E28	SCE/DESTN 6	45					
E29	SCE/DESTN 7	46	0.0	-150.0	-60.0	mA	1-4
E30	MAR/MWD 8	55	0.0	-260.0	-30.0	mA	1-4
E31	MAR/MWD 8	55					
E32	MAR/MWD 9	56					
E33	MAR/MWD 9	56					
E34	MAR/MWD 10	57					
E35	MAR/MWD 10	57					
E36	MAR/MWD 11	58					
E37	MAR/MWD 11	58					
E38	MAR/MWD 12	59					
E39	MAR/MWD 12	59					
E40	MAR/MWD 13	61	0.0	-260.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	MAR/MWD 13	61	0.0	-260.0	-30.0	mA	1-4
E42	MAR/MWD 14	62					
E43	MAR/MWD 14	62					
E44	MAR/MWD 15	63					
E45	MAR/MWD 15	63	0.0	-260.0	-30.0	mA	1-4
E46	IOC 11	68	0.0	-150.0	-60.0	mA	1-4
E47	IOC 10	69					
E48	IOC 9	70					
E49	IOC 8	71	0.0	-150.0	-60.0	mA	1-4
E50	SCE/DESTN 11	72	0.0	-220.0	-30.0	mA	1-4
E51	SCE/DESTN 11	72	0.0	-150.0	-60.0	mA	1-4
E52	SCE/DESTN 10	73	0.0	-220.0	-30.0	mA	1-4
E53	SCE/DESTN 10	73	0.0	-150.0	-60.0	mA	1-4
E54	SCE/DESTN 9	74	0.0	-220.0	-30.0	mA	1-4
E55	SCE/DESTN 9	74	0.0	-150.0	-60.0	mA	1-4
E56	SCE/DESTN 8	76	0.0	-220.0	-30.0	mA	1-4
E57	SCE/DESTN 8	76	0.0	-150.0	-60.0	mA	1-4
E58	IOC 15	82					
E59	IOC 14	83					
E60	IOC 13	85	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E61	IOC 12	86	0.0	-150.0	-60.0	mA	1-4
E62	SCE/DESTN 15	87	0.0	-220.0	-30.0	mA	1-4
E63	SCE/DESTN 15	87	0.0	-150.0	-60.0	mA	1-4
E64	SCE/DESTN 14	88	0.0	-220.0	-30.0	mA	1-4
E65	SCE/DESTN 14	88	0.0	-150.0	-60.0	mA	1-4
E66	SCE/DESTN 13	89	0.0	-220.0	-30.0	mA	1-4
E67	SCE/DESTN 13	89	0.0	-150.0	-60.0	mA	1-4
E68	SCE/DESTN 12	91	0.0	-220.0	-30.0	mA	1-4
E69	SCE/DESTN 12	91	0.0	-150.0	-60.0	mA	1-4
E70	RD DATA 8	93	0.0	-325.0	-100.0	mA	1-4
E71	RD DATA 9	94					
E72	RD DATA 10	95					
E73	RD DATA 11	96					
E74	RD DATA 12	97					
E75	RD DATA 13	98					
E76	RD DATA 14	99					
E77	RD DATA 15	100	0.0	-325.0	-100.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E77
Power supply current limit	1.44 A	NA	E1-E77
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E77
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E77 E1-E77
Output current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 8.75 mA ± 11.5 mA ± 8.75 mA ± 11.5 mA ± 8.75 mA ± 4.5 mA ± 11.5 mA ± 4.5 mA ± 9.5 mA ± 4.5 mA ± 9.5 mA ± 4.5 mA ± 9.5 mA ± 4.5 mA ± 9.5 mA ± 4.5 mA ± 9.5 mA ± 4.5 mA ± 9.5 mA ± 4.5 mA ± 9.5 mA ± 4.5 mA	E1 E2 E3,E4 E5-E20 E21 E22-E29 E30-E45 E46-E49 E50 E51 E52 E53 E54 E55 E56 E57-E61 E62 E63 E64 E65 E66 E67

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA NA NA	± 9.5 mA ± 4.5 mA ± 11.2 mA	E68 E69 E70-E77

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	IO CLK	2	2.4	-14.0	0.35	mA	1-3
F2	ENBL CIII	5	2.4	-2.0	0.05	mA	1-3
F3	IOC LP BK	6	2.4	-0.6	0.02	mA	1-3
F4	POWER MACLR	7					
F5	CP MACLR	8	2.4	-0.6	0.02	mA	1-3
F6	DESTN MDR	9	2.4	-1.8	0.06	mA	1-3
F7	DESTN MAR	12	2.4	-2.4	0.08	mA	1-3
F8	MPC MACLR	13	2.4	-0.6	0.02	mA	1-3
F9	DUAL 1ST	15	2.4	-0.6	0.02	mA	1-3
F10	CLIII + T4	17	2.4	-1.4	0.06	mA	1-3
F11	ENBL T1	18	2.4	-2.6	0.1	mA	1-3
F12	ENBL MDRO	26	2.4	-0.4	0.04	mA	1-3
F13	20 MHZ	28	2.4	-12.0	0.3	mA	1-3
F14	SEL T4	32	2.4	-2.4	0.08	mA	1-3
F15	ENBL MAR	33	2.4	-0.4	0.04	mA	1-3
F16	SCE/DESTN 0	34	2.4	-0.65	0.07	mA	1-3
F17	SCE/DESTN 1	35					
F18	SCE/DESTN 2	36					
F19	SCE/DESTN 3	37	2.4	-0.65	0.07	mA	1-3
F20	IOC 0	38	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F21	IOC 1	39	2.4	-0.6	0.02	mA	1-3
F22	IOC 2	41					
F23	IOC 3	42	2.4	-0.6	0.02	mA	1-3
F24	SCE/DESTN 4	43	2.4	-0.65	0.07	mA	1-3
F25	SCE/DESTN 5	44					
F26	SCE/DESTN 6	45					
F27	SCE/DESTN 7	46	2.4	-0.65	0.07	mA	1-3
F28	IOC 4	47	2.4	-0.6	0.02	mA	1-3
F29	IOC 5	48					
F30	IOC 6	49					
F31	IOC 7	50					
F32	S/D LTCH	52	2.4	-0.6	0.02	mA	1-3
F33	RD DTR-CLK	53	2.4	-0.2	0.02	mA	1-3
F34	MDR + T3	54	2.4	-1.2	0.04	mA	1-3
F35	MEM PTCT	64	2.4	-1.2	0.04	mA	1-3
F36	I/O INST FLT	65	2.4	-3.2	0.09	mA	1-3
F37	CLK CII	66	2.4	-0.2	0.02	mA	1-3
F38	ENBL INTR	67	2.4	-2.6	0.06	mA	1-3
F39	IOC 11	68	2.4	-0.65	0.07	mA	1-3
F40	IOC 10	69	2.4	-0.65	0.07	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	IOC 9	70	2.4	-0.65	0.07	mA	1-3
F42	IOC 8	71	2.4	-0.65	0.07	mA	1-3
F43	SCE/DESTN 11	72	2.4	-0.87	0.11	mA	1-3
F44	SCE/DESTN 10	73					
F45	SCE/DESTN 9	74					
F46	SCE/DESTN 8	76	2.4	-0.87	0.11	mA	1-3
F47	ADRS 1	77	2.4	-0.6	0.02	mA	1-3
F48	CLK CI	78	2.4	-0.2	0.02	mA	1-3
F49	PARITY ER	79	2.4	-0.8	0.04	mA	1-3
F50	DATA/ADRS TO	80	2.4	-0.8	0.04	mA	1-3
F51	IOC 15	82	2.4	-0.65	0.07	mA	1-3
F52	IOC 14	83	2.4	-0.65	0.07	mA	1-3
F53	ADRS 0	84	2.4	-0.6	0.02	mA	1-3
F54	IOC 13	85	2.4	-0.65	0.07	mA	1-3
F55	IOC 12	86	2.4	-0.65	0.07	mA	1-3
F56	SCE/DESTN 15	87	2.4	-0.87	0.11	mA	1-3
F57	SCE/DESTN 14	88					
F58	SCE/DESTN 13	89					
F59	SCE/DESTN 12	91	2.4	-0.87	0.11	mA	1-3
F60	MPC MODE	92	2.4	-5.2	0.14	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F61	RD DATA 8	93	2.4	-0.65	0.07	mA	1-3
F62	RD DATA 9	94					
F63	RD DATA 10	95					
F64	RD DATA 11	96					
F65	RD DATA 12	97					
F66	RD DATA 13	98					
F67	RD DATA 14	99					
F68	RD DATA 15	100	2.4	-0.65	0.07	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F68
Power supply current limit	1.44 A	NA	F1-F68
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F68
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F68
Pattern voltage low, (VL)	0.0 V	± 100 mV	F1-F68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 720 uA	F1
	NA	± 102 uA	F2
	NA	± 31 uA	F3-F5
	NA	± 93 uA	F6
	NA	± 124 uA	F7
	NA	± 31 uA	F8,F9
	NA	± 73 uA	F10
	NA	± 135 uA	F11
	NA	± 22 uA	F12
	NA	± 620 uA	F13
	NA	± 124 uA	F14
	NA	± 22 uA	F15
	NA	± 36 uA	F16-F19
	NA	± 31 uA	F20-F23
	NA	± 36 uA	F24-F27
	NA	± 31 uA	F28-F32
	NA	± 11 uA	F33
	NA	± 62 uA	F34,F35
	NA	± 164 uA	F36
	NA	± 11 uA	F37
	NA	± 133 uA	F38
	NA	± 36 uA	F39-F42
	NA	± 49 uA	F43-F46
	NA	± 31 uA	F47
	NA	± 11 uA	F48
	NA	± 42 uA	F49,F50
	NA	± 36 uA	F51,F52
	NA	± 31 uA	F53
	NA	± 36 uA	F54,F55
	NA	± 49 uA	F56-F59
	NA	± 267 uA	F60
	NA	± 36 uA	F61-F68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	IO CLK	2	0.4	-14.0	0.35	mA	1-3
G2	ENBL CIII	5	0.4	-2.0	0.05	mA	1-3
G3	IOC LP BK	6	0.4	-0.6	0.02	mA	1-3
G4	POWER MACLR	7					
G5	CP MACLR	8	0.4	-0.6	0.02	mA	1-3
G6	DESTN MDR	9	0.4	-1.8	0.06	mA	1-3
G7	DESTN MAR	12	0.4	-2.4	0.08	mA	1-3
G8	MPC MACLR	13	0.4	-0.6	0.02	mA	1-3
G9	DUAL 1ST	15	0.4	-0.6	0.02	mA	1-3
G10	CLIII + T4	17	0.4	-1.4	0.06	mA	1-3
G11	ENBL T1	18	0.4	-2.6	0.1	mA	1-3
G12	ENBL MDRO	26	0.4	-0.4	0.04	mA	1-3
G13	20 MHZ	28	0.4	-12.0	0.3	mA	1-3
G14	SEL T4	32	0.4	-2.4	0.08	mA	1-3
G15	ENBL MAR	33	0.4	-0.4	0.04	mA	1-3
G16	SCE/DESTN 0	34	0.4	-0.65	0.07	mA	1-3
G17	SCE/DESTN 1	35					
G18	SCE/DESTN 2	36					
G19	SCE/DESTN 3	37	0.4	-0.65	0.07	mA	1-3
G20	IOC 0	38	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	IOC 1	39	0.4	-0.6	0.02	mA	1-3
G22	IOC 2	41					
G23	IOC 3	42	0.4	-0.6	0.02	mA	1-3
G24	SCE/DESTN 4	43	0.4	-0.65	0.07	mA	1-3
G25	SCE/DESTN 5	44					
G26	SCE/DESTN 6	45					
G27	SCE/DESTN 7	46	0.4	-0.65	0.07	mA	1-3
G28	IOC 4	47	0.4	-0.6	0.02	mA	1-3
G29	IOC 5	48					
G30	IOC 6	49					
G31	IOC 7	50					
G32	S/D LTCH	52	0.4	-0.6	0.02	mA	1-3
G33	RD DTR-CLK	53	0.4	-0.2	0.02	mA	1-3
G34	MDR + T3	54	0.4	-1.2	0.04	mA	1-3
G35	MEM PTCT	64	0.4	-1.2	0.04	mA	1-3
G36	I/O INST FLT	65	0.4	-3.2	0.09	mA	1-3
G37	CLK CII	66	0.4	-0.2	0.02	mA	1-3
G38	ENBL INTR	67	0.4	-2.6	0.06	mA	1-3
G39	IOC 11	68	0.4	-0.65	0.07	mA	1-3
G40	IOC 10	69	0.4	-0.65	0.07	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	IOC 9	70	0.4	-0.65	0.07	mA	1-3
G42	IOC 8	71	0.4	-0.65	0.07	mA	1-3
G43	SCE/DESTN 11	72	0.4	-0.87	0.11	mA	1-3
G44	SCE/DESTN 10	73					
G45	SCE/DESTN 9	74					
G46	SCE/DESTN 8	76	0.4	-0.87	0.11	mA	1-3
G47	ADRS 1	77	0.4	-0.6	0.02	mA	1-3
G48	CLK CI	78	0.4	-0.2	0.02	mA	1-3
G49	PARITY ER	79	0.4	-0.8	0.04	mA	1-3
G50	DATA/ADRS TO	80	0.4	-0.8	0.04	mA	1-3
G51	IOC 15	82	0.4	-0.65	0.07	mA	1-3
G52	IOC 14	83	0.4	-0.65	0.07	mA	1-3
G53	ADRS 0	84	0.4	-0.6	0.02	mA	1-3
G54	IOC 13	85	0.4	-0.65	0.07	mA	1-3
G55	IOC 12	86	0.4	-0.65	0.07	mA	1-3
G56	SCE/DESTN 15	87	0.4	-0.87	0.11	mA	1-3
G57	SCE/DESTN 14	88					
G58	SCE/DESTN 13	89					
G59	SCE/DESTN 12	91	0.4	-0.87	0.11	mA	1-3
G60	MPC MODE	92	0.4	-5.2	0.14	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G61	RD DATA 8	93	0.4	-0.65	0.07	mA	1-3
G62	RD DATA 9	94					
G63	RD DATA 10	95					
G64	RD DATA 11	96					
G65	RD DATA 12	97					
G66	RD DATA 13	98					
G67	RD DATA 14	99					
G68	RD DATA 15	100	0.4	-0.65	0.07	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G68
Power supply current limit	1.44 A	NA	G1-G68
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G68
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G68
low, (VL)	0.0 V	± 100 mV	G1-G68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 720 uA	G1
	NA	± 102 uA	G2
	NA	± 31 uA	G3-G5
	NA	± 93 uA	G6
	NA	± 124 uA	G7
	NA	± 31 uA	G8,G9
	NA	± 73 uA	G10
	NA	± 135 uA	G11
	NA	± 22 uA	G12
	NA	± 620 uA	G13
	NA	± 124 uA	G14
	NA	± 22 uA	G15
	NA	± 36 uA	G16-G19
	NA	± 31 uA	G20-G23
	NA	± 36 uA	G24-G27
	NA	± 31 uA	G28-G32
	NA	± 11 uA	G33
	NA	± 62 uA	G34,G35
	NA	± 164 uA	G36
	NA	± 11 uA	G37
	NA	± 133 uA	G38
	NA	± 36 uA	G39-G42
	NA	± 49 uA	G43-G46
	NA	± 31 uA	G47
	NA	± 11 uA	G48
	NA	± 42 uA	G49,G50
	NA	± 36 uA	G51,G52
	NA	± 31 uA	G53
	NA	± 36 uA	G54,G55
	NA	± 49 uA	G56-G59
	NA	± 267 uA	G60
	NA	± 36 uA	G61-G68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	IOC MACLR	3	-0.4	2.4	4.5	V	1-4
H2	MDRI 9	4	-0.16	2.4	4.5	V	1-4
H3	CLK B	11	-0.4	2.4	4.5	V	1-4
H4	CLK A	14	-0.4	2.4	4.5	V	1-4
H5	MAR/MWD 0	16	-0.16	2.4	4.5	V	1-4
H6	MAR/MWD 0	16					
H7	MAR/MWD 1	19					
H8	MAR/MWD 1	19					
H9	MAR/MWD 2	20					
H10	MAR/MWD 2	20					
H11	MAR/MWD 3	21					
H12	MAR/MWD 3	21					
H13	MAR/MWD 4	22					
H14	MAR/MWD 4	22					
H15	MAR/MWD 5	23					
H16	MAR/MWD 5	23					
H17	MAR/MWD 6	24					
H18	MAR/MWD 6	24					
H19	MAR/MWD 7	27					
H20	MAR/MWD 7	27	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	2.457 MHZ	29	-0.4	2.4	4.5	V	1-4
H22	SCE/DESTN 0	34					
H23	SCE/DESTN 1	35					
H24	SCE/DESTN 2	36					
H25	SCE/DESTN 3	37					
H26	SCE/DESTN 4	43					
H27	SCE/DESTN 5	44					
H28	SCE/DESTN 6	45					
H29	SCE/DESTN 7	46	-0.4	2.4	4.5	V	1-4
H30	MAR/MWD 8	55	-0.16	2.4	4.5	V	1-4
H31	MAR/MWD 8	55					
H32	MAR/MWD 9	56					
H33	MAR/MWD 9	56					
H34	MAR/MWD 10	57					
H35	MAR/MWD 10	57					
H36	MAR/MWD 11	58					
H37	MAR/MWD 11	58					
H38	MAR/MWD 12	59					
H39	MAR/MWD 12	59					
H40	MAR/MWD 13	61	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	MAR/MWD 13	61	-0.16	2.4	4.5	V	1-4
H42	MAR/MWD 14	62					
H43	MAR/MWD 14	62					
H44	MAR/MWD 15	63					
H45	MAR/MWD 15	63	-0.16	2.4	4.5	V	1-4
H46	IOC 11	68	-0.4	2.4	4.5	V	1-4
H47	IOC 10	69					
H48	IOC 9	70					
H49	IOC 8	71	-0.4	2.4	4.5	V	1-4
H50	SCE/DESTN 11	72	-0.16	2.4	4.5	V	1-4
H51	SCE/DESTN 11	72					
H52	SCE/DESTN 10	73					
H53	SCE/DESTN 10	73					
H54	SCE/DESTN 9	74					
H55	SCE/DESTN 9	74					
H56	SCE/DESTN 8	76					
H57	SCE/DESTN 8	76	-0.16	2.4	4.5	V	1-4
H58	IOC 15	82	-0.4	2.4	4.5	V	1-4
H59	IOC 14	83					
H60	IOC 13	85	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H61	IOC 12	86	-0.4	2.4	4.5	V	1-4
H62	SCE/DESTN 15	87	-0.16	2.4	4.5	V	1-4
H63	SCE/DESTN 15	87					
H64	SCE/DESTN 14	88					
H65	SCE/DESTN 14	88					
H66	SCE/DESTN 13	89					
H67	SCE/DESTN 13	89					
H68	SCE/DESTN 12	91					
H69	SCE/DESTN 12	91	-0.16	2.4	4.5	V	1-4
H70	RD DATA 8	93	-0.4	2.4	4.5	V	1-4
H71	RD DATA 9	94					
H72	RD DATA 10	95					
H73	RD DATA 11	96					
H74	RD DATA 12	97					
H75	RD DATA 13	98					
H76	RD DATA 14	99					
H77	RD DATA 15	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H77
Power supply current limit	1.44 A	NA	H1-H77
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1
	-0.16 mA	± 16 uA	H2
	-0.4 mA	± 40 uA	H3,H4
	-0.16 mA	± 16 uA	H5-H20
	-0.4 mA	± 40 uA	H21-H29
	-0.16 mA	± 16 uA	H30-H45
	-0.4 mA	± 40 uA	H46-H49
	-0.16 mA	± 16 uA	H50-H57
	-0.4 mA	± 40 uA	H58-H61
	-0.16 mA	± 16 uA	H62-H69
	-0.4 mA	± 40 uA	H70-H77
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H77
low, (VL)	3.0 V	± 100 mV	H1-H77
Output voltage measurement	NA	± 105 mV	H1-H77

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

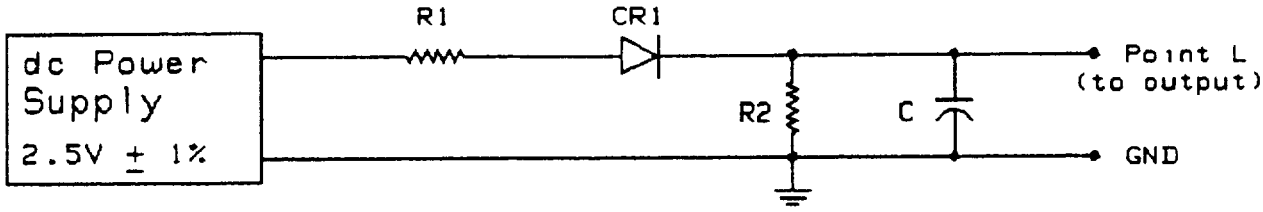
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
3,11,14,29,34,35, 36,37,43,44,45,46, 68,69,70,71,82,83, 85,86,93,94,95,96, 97,98	6200	6.0	0.5
99,100	6200	NA	NA
4,16,19,20,21,22, 23,24,27,55,56,57, 58,59,61,62,63,72, 73,74,76,87,88,89, 91	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	IOC MACLR	3	6.0	0.0	0.5	V	1-4
J2	MDRI 9	4	1.6	0.0	0.5	V	1-4
J3	CLK B	11	6.0	0.0	0.5	V	1-4
J4	CLK A	14	6.0	0.0	0.5	V	1-4
J5	MAR/MWD 0	16	1.6	0.0	0.5	V	1-4
J6	MAR/MWD 0	16					
J7	MAR/MWD 1	19					
J8	MAR/MWD 1	19					
J9	MAR/MWD 2	20					
J10	MAR/MWD 2	20					
J11	MAR/MWD 3	21					
J12	MAR/MWD 3	21					
J13	MAR/MWD 4	22					
J14	MAR/MWD 4	22					
J15	MAR/MWD 5	23					
J16	MAR/MWD 5	23					
J17	MAR/MWD 6	24					
J18	MAR/MWD 6	24					
J19	MAR/MWD 7	27					
J20	MAR/MWD 7	27	1.6	0.0	0.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	2.457 MHZ	29	6.0	0.0	0.5	V	1-4
J22	SCE/DESTN 0	34					
J23	SCE/DESTN 1	35					
J24	SCE/DESTN 2	36					
J25	SCE/DESTN 3	37					
J26	SCE/DESTN 4	43					
J27	SCE/DESTN 5	44					
J28	SCE/DESTN 6	45					
J29	SCE/DESTN 7	46	6.0	0.0	0.5	V	1-4
J30	MAR/MWD 8	55	1.6	0.0	0.5	V	1-4
J31	MAR/MWD 8	55					
J32	MAR/MWD 9	56					
J33	MAR/MWD 9	56					
J34	MAR/MWD 10	57					
J35	MAR/MWD 10	57					
J36	MAR/MWD 11	58					
J37	MAR/MWD 11	58					
J38	MAR/MWD 12	59					
J39	MAR/MWD 12	59					
J40	MAR/MWD 13	61	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	MAR/MWD 13	61	1.6	0.0	0.5	V	1-4
J42	MAR/MWD 14	62					
J43	MAR/MWD 14	62					
J44	MAR/MWD 15	63					
J45	MAR/MWD 15	63	1.6	0.0	0.5	V	1-4
J46	IOC 11	68	6.0	0.0	0.5	V	1-4
J47	IOC 10	69					
J48	IOC 9	70					
J49	IOC 8	71	6.0	0.0	0.5	V	1-4
J50	SCE/DESTN 11	72	1.6	0.0	0.5	V	1-4
J51	SCE/DESTN 11	72					
J52	SCE/DESTN 10	73					
J53	SCE/DESTN 10	73					
J54	SCE/DESTN 9	74					
J55	SCE/DESTN 9	74					
J56	SCE/DESTN 8	76					
J57	SCE/DESTN 8	76	1.6	0.0	0.5	V	1-4
J58	IOC 15	82	6.0	0.0	0.5	V	1-4
J59	IOC 14	83					
J60	IOC 13	85	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J61	IOC 12	86	6.0	0.0	0.5	V	1-4
J62	SCE/DESTN 15	87	1.6	0.0	0.5	V	1-4
J63	SCE/DESTN 15	87					
J64	SCE/DESTN 14	88					
J65	SCE/DESTN 14	88					
J66	SCE/DESTN 13	89					
J67	SCE/DESTN 13	89					
J68	SCE/DESTN 12	91					
J69	SCE/DESTN 12	91	1.6	0.0	0.5	V	1-4
J70	RD DATA 8	93	6.0	0.0	0.5	V	1-4
J71	RD DATA 9	94					
J72	RD DATA 10	95					
J73	RD DATA 11	96					
J74	RD DATA 12	97					
J75	RD DATA 13	98	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:38Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J75
Power supply current limit	1.44 A	NA	J1-J75
Forcing current, (Iol)	6.0 mA	± 600 uA	J1
	1.6 mA	± 160 uA	J2
	6.0 mA	± 600 uA	J3,J4
	1.6 mA	± 160 uA	J5-J20
	6.0 mA	± 600 uA	J21-J29
	1.6 mA	± 160 uA	J30-J45
	6.0 mA	± 600 uA	J46-J49
	1.6 mA	± 160 uA	J50-J57
	6.0 mA	± 600 uA	J58-J61
	1.6 mA	± 160 uA	J62-J69
	6.0 mA	± 600 uA	J70-J75
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J75
low, (VL)	3.0 V	± 100 mV	J1-J75
Output voltage measurement	NA	± 25 mV	J1-J75

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	MAR/MWD 0	16	2.4	-0.04	0.04	mA	1-3
K2	MAR/MWD 1	19					
K3	MAR/MWD 2	20					
K4	MAR/MWD 3	21					
K5	MAR/MWD 4	22					
K6	MAR/MWD 5	23					
K7	MAR/MWD 6	24					
K8	MAR/MWD 7	27					
K9	MAR/MWD 8	55					
K10	MAR/MWD 9	56					
K11	MAR/MWD 10	57					
K12	MAR/MWD 11	58					
K13	MAR/MWD 12	59					
K14	MAR/MWD 13	61					
K15	MAR/MWD 14	62					
K16	MAR/MWD 15	63	2.4	-0.04	0.04	mA	1-3
Low level							
K17	MAR/MWD 0	16	0.4	-0.04	0.04	mA	1-3
K18	MAR/MWD 1	19					
K19	MAR/MWD 2	20	0.4	-0.04	0.04	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K20	MAR/MWD 3	21	0.4	-0.04	0.04	mA	1-3
K21	MAR/MWD 4	22					
K22	MAR/MWD 5	23					
K23	MAR/MWD 6	24					
K24	MAR/MWD 7	27					
K25	MAR/MWD 8	55					
K26	MAR/MWD 9	56					
K27	MAR/MWD 10	57					
K28	MAR/MWD 11	58					
K29	MAR/MWD 12	59					
K30	MAR/MWD 13	61					
K31	MAR/MWD 14	62					
K32	MAR/MWD 15	63	0.4	-0.04	0.04	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K32
Power supply current limit	1.44 A	NA	K1-K32
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K16 K17-K32
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K32 K1-K32
Output current measurement	NA	± 4 uA	K1-K32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	IO CLK	2	-18.0	-1.2	0.0	V	1-3
L2	ENBL CIII	5					
L3	IOC LP BK	6					
L4	POWER MACLR	7					
L5	CP MACLR	8					
L6	DESTN MDR	9					
L7	DESTN MAR	12					
L8	MPC MACLR	13					
L9	DUAL 1ST	15					
L10	CLIII + T4	17					
L11	ENBL T1	18	-18.0	-1.2	0.0	V	1-3
L12	ENBL MDRO	26	-18.0	-1.5	0.0	V	1-3
L13	20 MHZ	28	-18.0	-1.2	0.0	V	1-3
L14	SEL T4	32	-18.0	-1.2	0.0	V	1-3
L15	ENBL MAR	33	-18.0	-1.5	0.0	V	1-3
L16	IOC 0	38	-18.0	-1.2	0.0	V	1-3
L17	IOC 1	39					
L18	IOC 2	41					
L19	IOC 3	42					
L20	IOC 4	47	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	IOC 5	48	-18.0	-1.2	0.0	V	1-3
L22	IOC 6	49					
L23	IOC 7	50					
L24	S/D LTCH	52	-18.0	-1.2	0.0	V	1-3
L25	RD DTR-CLK	53	-18.0	-1.5	0.0	V	1-3
L26	MDR + T3	54	-18.0	-1.2	0.0	V	1-3
L27	MEM PTCT	64					
L28	I/O INST FLT	65	-18.0	-1.2	0.0	V	1-3
L29	CLK CII	66	-18.0	-1.5	0.0	V	1-3
L30	ENBL INTR	67	-18.0	-1.2	0.0	V	1-3
L31	ADRS 1	77	-18.0	-1.2	0.0	V	1-3
L32	CLK CI	78	-18.0	-1.5	0.0	V	1-3
L33	PARITY ER	79	-18.0	-1.2	0.0	V	1-3
L34	DATA/ADRS TO	80					
L35	ADRS 0	84					
L36	MPC MODE	92	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L36
Power supply current limit	NA	NA	L1-L36
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L36
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L36 L1-L36
Input voltage measurement	NA NA NA NA NA NA NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1-L11 L12 L13,L14 L15 L16-L24 L25 L26-L28 L29 L30,L31 L32 L33-L36

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	5.6	10.4	uF	1-4
M2	PSFC	51	6.3	11.7	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 240 nF ± 270 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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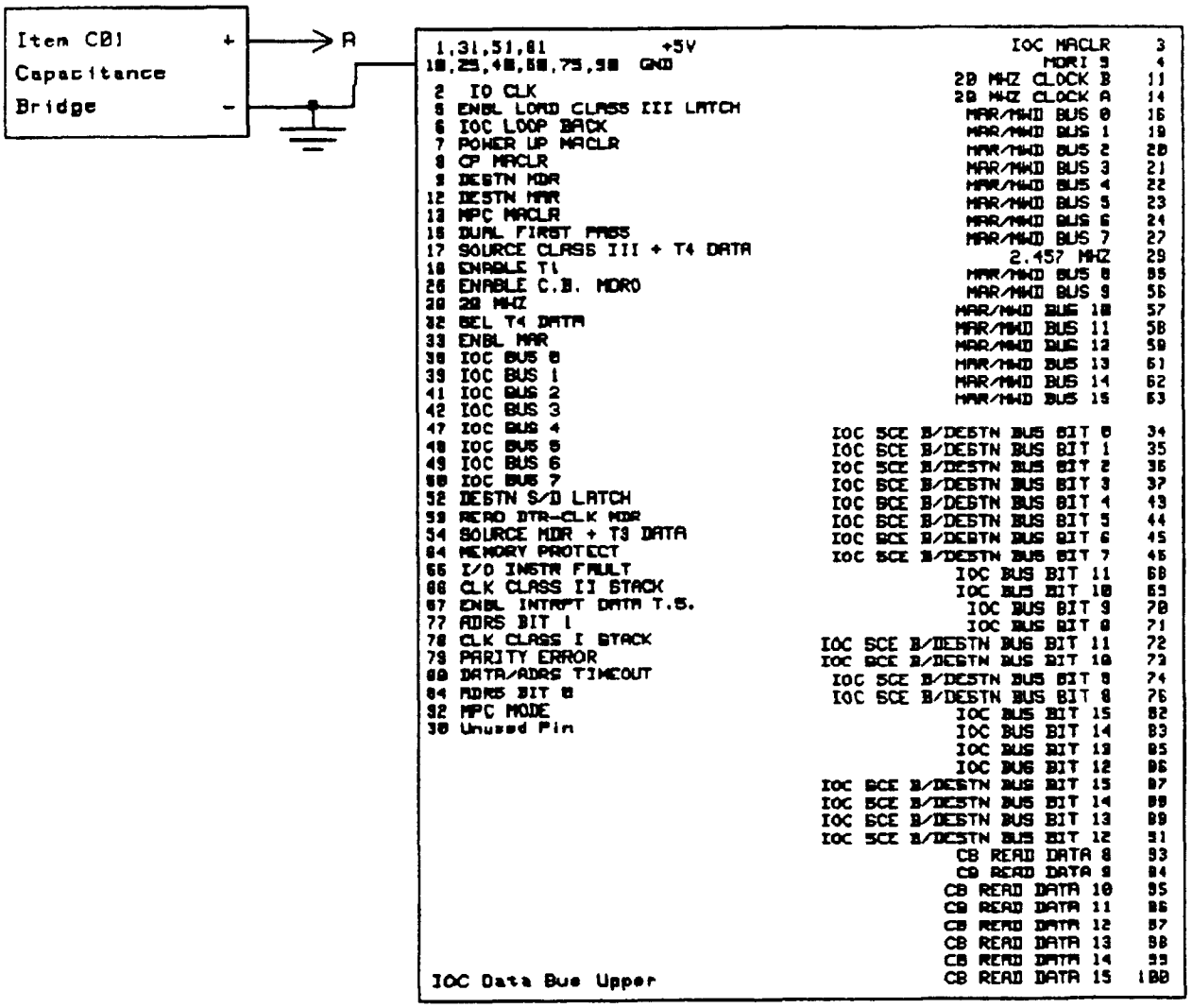


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	400	960	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.44 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 28 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.44 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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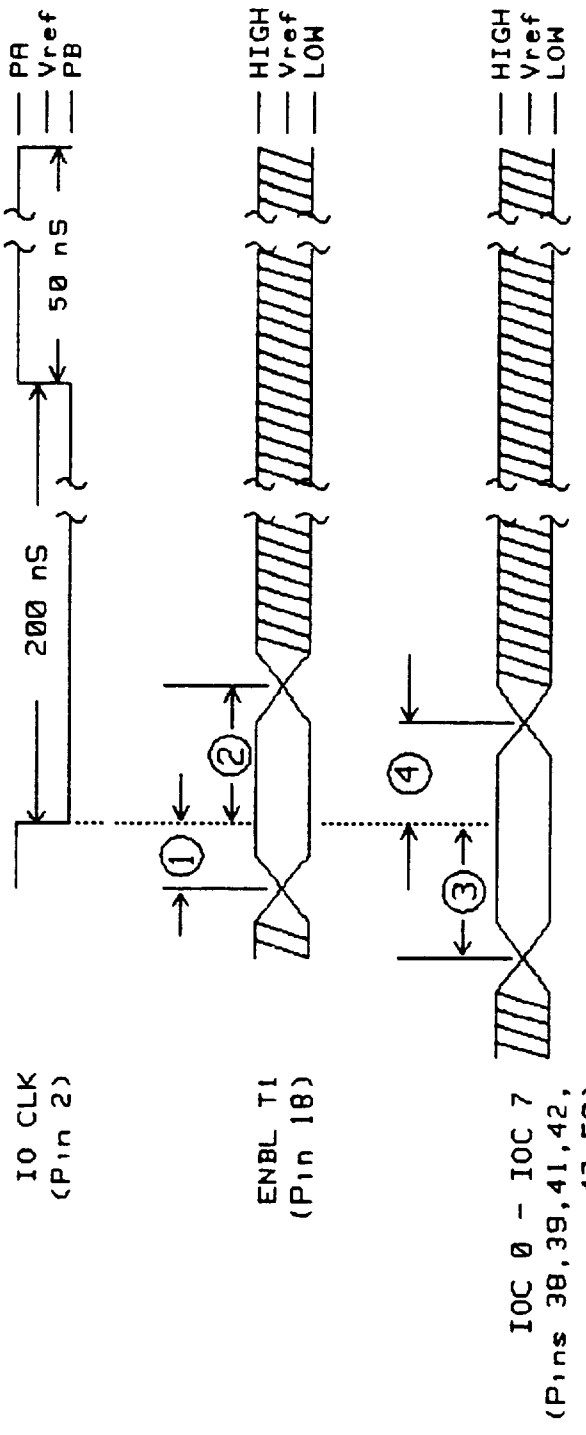


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) ENBL T1 to IO CLK. Setup time: 12 nS minimum.
2. (2) IO CLK to ENBL T1. Hold time: 14 nS minimum.
3. (3) IOC 0 - IOC 7 to IO CLK. Setup time: 24 nS minimum.
4. (4) IO CLK to IOC 0 - IOC 7. Hold time: 11 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.44 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

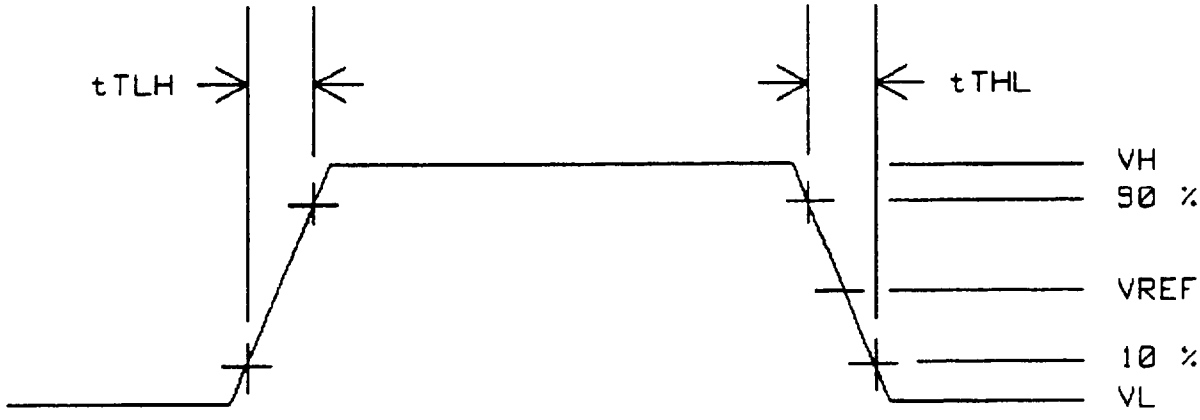
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CLIII + T4 to SCE/DESTN 0	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 1	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 2	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 3	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 4	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 5	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 6	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 7	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 8	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 9	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 10	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 11	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 12	-	-	-	32	32	32	nS	12a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CLIII + T4 to SCE/DESTN 13	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 14	-	-	-	32	32	32	nS	12a
1	tpd	CLIII + T4 to SCE/DESTN 15	-	-	-	32	32	32	nS	12a
2	tpd	IO CLK to MAR/MWD 0	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 1	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 2	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 3	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 4	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 5	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 6	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 7	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 8	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 9	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 10	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 11	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 12	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 13	-	-	-	36	36	36	nS	12a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	IO CLK to MAR/MWD 14	-	-	-	36	36	36	nS	12a
2	tpd	IO CLK to MAR/MWD 15	-	-	-	36	36	36	nS	12a
3	tpd	RD DTR-CLK to MDRI 9	-	-	-	26	26	26	nS	12a
4	tpd	IO CLK to IOC 8	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 9	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 10	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 11	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 12	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 13	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 14	-	-	-	28	28	28	nS	12a
4	tpd	IO CLK to IOC 15	-	-	-	28	28	28	nS	12a
5	tpd	IO CLK to RD DATA 8	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 9	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 10	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 11	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 12	-	-	-	51	51	51	nS	12a
5	tpd	IO CLK to RD DATA 13	-	-	-	51	51	51	nS	12a
6	tsu	ENBL T1 to IO CLK	12	12	12	-	-	-	nS	12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	th	IO CLK to ENBL T1	14	14	14	-	-	-	nS	12b
8	tsu	IOC 0 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 1 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 2 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 3 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 4 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 5 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 6 to IO CLK	24	24	24	-	-	-	nS	12b
8	tsu	IOC 7 to IO CLK	24	24	24	-	-	-	nS	12b
9	th	IO CLK to IOC 0	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 1	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 2	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 3	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 4	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 5	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 6	11	11	11	-	-	-	nS	12b
9	th	IO CLK to IOC 7	11	11	11	-	-	-	nS	12b

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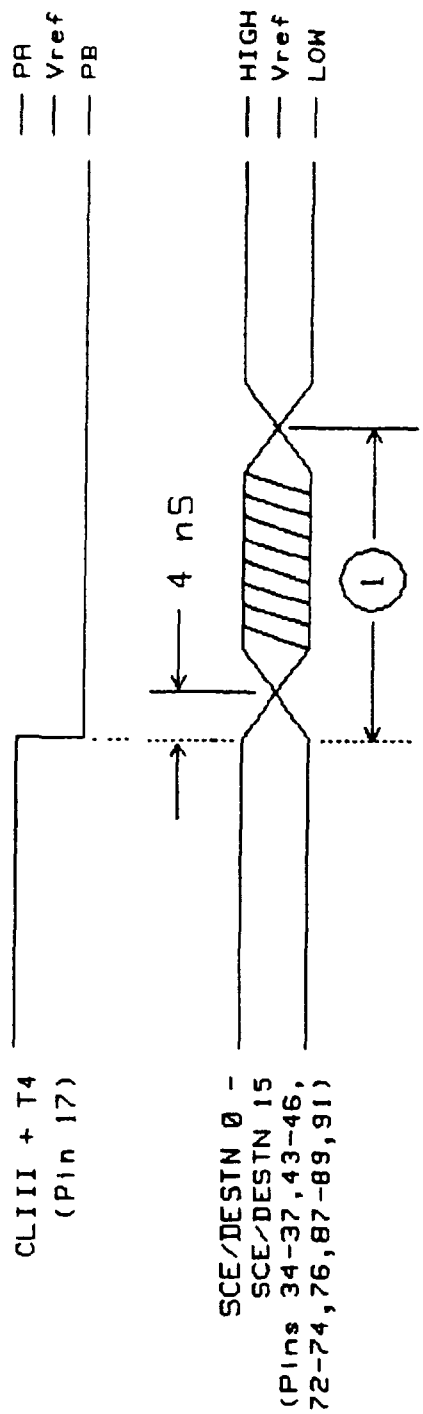


FIGURE 12a. Dynamic waveforms for table XVII.

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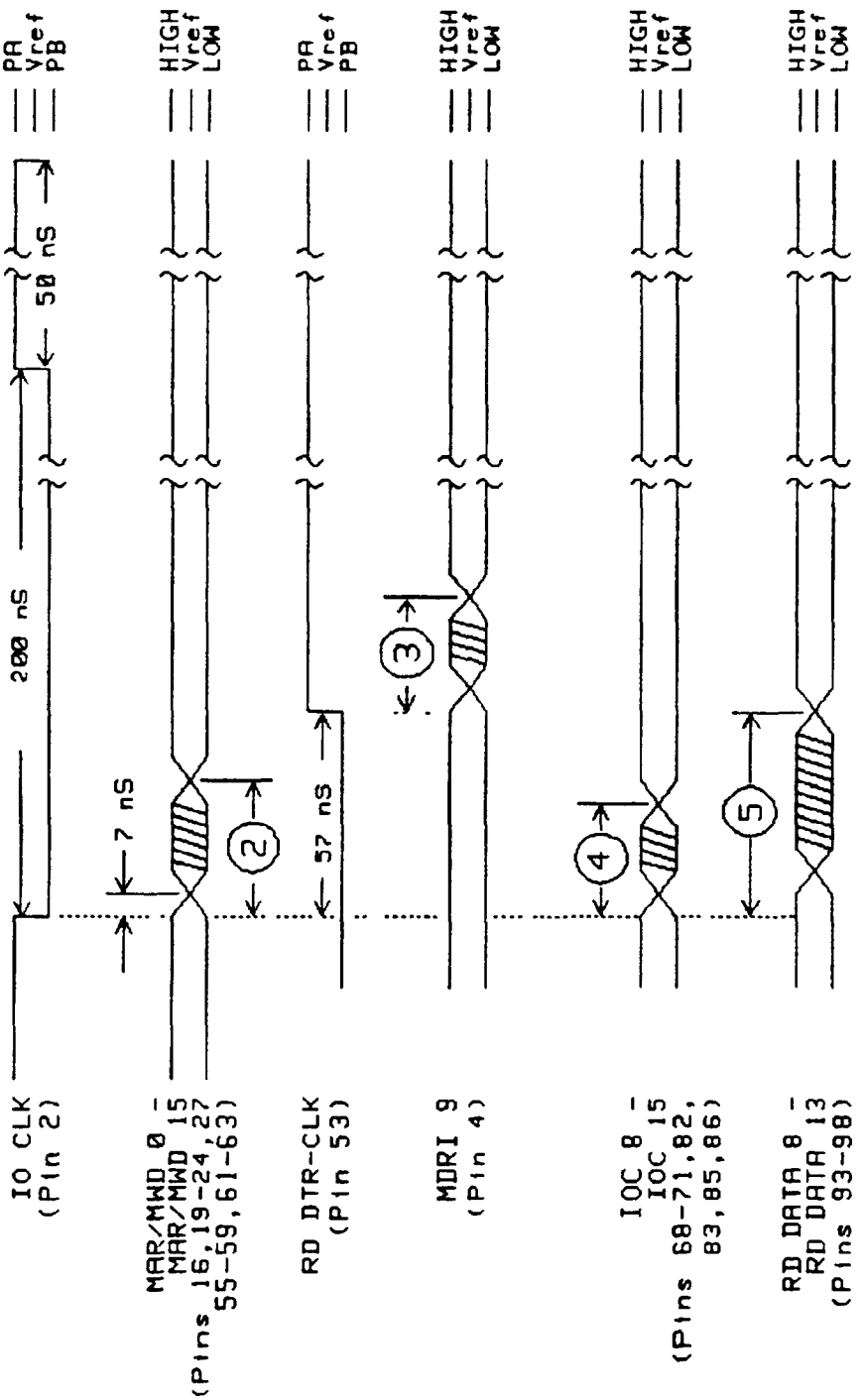


FIGURE 12b. Dynamic waveforms for table XVII.

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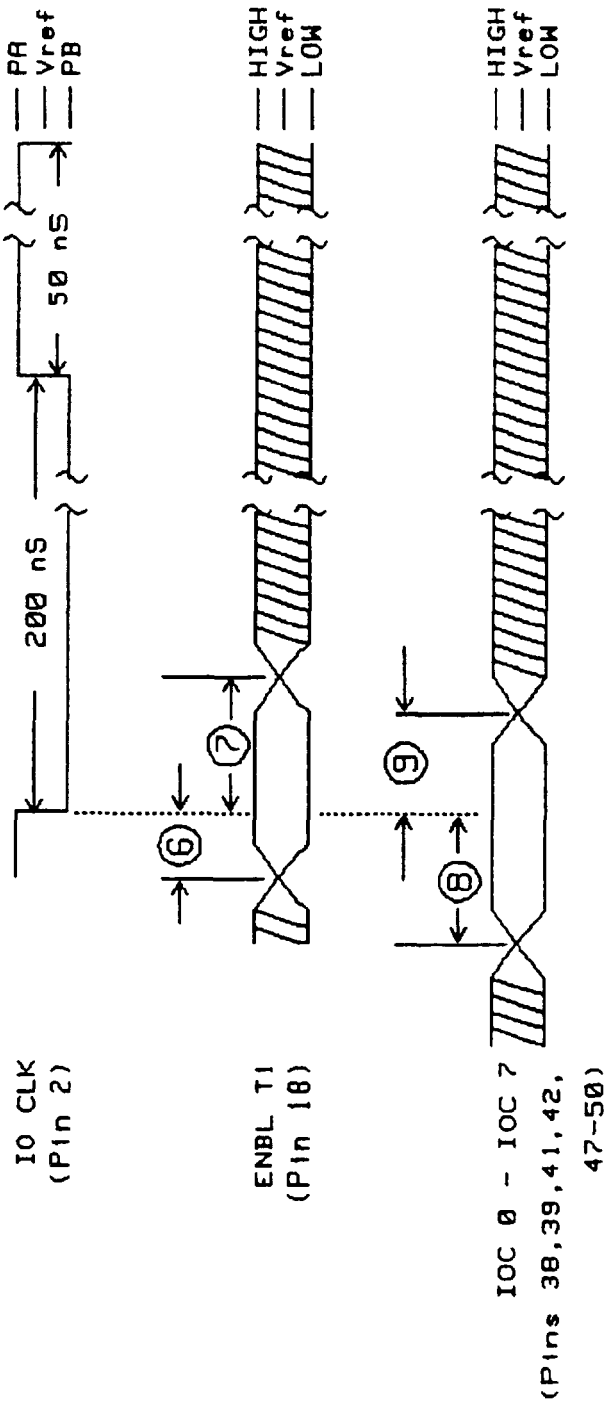


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	IO CLK	2	10	150	pF	1-4
R2	ENBL CIII	5	2	75	pF	1-4
R3	IOC LP BK	6	2	50	pF	1-4
R4	POWER MACLR	7	2	75	pF	1-4
R5	CP MACLR	8	2	75	pF	1-4
R6	DESTN MDR	9	6	75	pF	1-4
R7	DESTN MAR	12	8	100	pF	1-4
R8	MPC MACLR	13	2	75	pF	1-4
R9	DUAL 1ST	15	2	75	pF	1-4
R10	CLIII + T4	17	6	150	pF	1-4
R11	ENBL T1	18	10	150	pF	1-4
R12	ENBL MDRO	26	4	100	pF	1-4
R13	20 MHZ	28	6	100	pF	1-4
R14	SEL T4	32	6	150	pF	1-4
R15	ENBL MAR	33	4	100	pF	1-4
R16	IOC 0	38	2	75	pF	1-4
R17	IOC 1	39	2	50	pF	1-4
R18	IOC 2	41	2	75	pF	1-4
R19	IOC 3	42	|	|	|	|
R20	IOC 4	47	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	IOC 5	48	2	75	pF	1-4
R22	IOC 6	49				
R23	IOC 7	50	2	75	pF	1-4
R24	S/D LTCH	52	2	100	pF	1-4
R25	RD DTR-CLK	53	2	100	pF	1-4
R26	MDR + T3	54	4	75	pF	1-4
R27	MEM PTCT	64	4	100	pF	1-4
R28	I/O INST FLT	65	4	125	pF	1-4
R29	CLK CII	66	2	125	pF	1-4
R30	ENBL INTR	67	6	100	pF	1-4
R31	ADRS 1	77	2	75	pF	1-4
R32	CLK CI	78	2	75	pF	1-4
R33	PARITY ER	79	4	75	pF	1-4
R34	DATA/ADRS TO	80	4	75	pF	1-4
R35	ADRS 0	84	2	75	pF	1-4
R36	MPC MODE	92	2	150	pF	1-4
Outputs						
R37	MAR/MWD 0	16	4	100	pF	1-4
R38	MAR/MWD 1	19				
R39	MAR/MWD 2	20				
R40	MAR/MWD 3	21	4	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R41	MAR/MWD 4	22	4	100	pF	1-4
R42	MAR/MWD 5	23				
R43	MAR/MWD 6	24				
R44	MAR/MWD 7	27	4	100	pF	1-4
Bidirectionals						
R45	SCE/DESTN 0	34	4	75	pF	1-4
R46	SCE/DESTN 1	35				
R47	SCE/DESTN 2	36				
R48	SCE/DESTN 3	37				
R49	SCE/DESTN 4	43				
R50	SCE/DESTN 5	44				
R51	SCE/DESTN 6	45				
R52	SCE/DESTN 7	46	4	75	pF	1-4
Outputs						
R53	MAR/MWD 8	55	4	75	pF	1-4
R54	MAR/MWD 9	56				
R55	MAR/MWD 10	57				
R56	MAR/MWD 11	58				
R57	MAR/MWD 12	59				
R58	MAR/MWD 13	61				
R59	MAR/MWD 14	62	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R60	MAR/MWD 15	63	4	100	pF	1-4
Bidirectionals						
R61	IOC 11	68	4	75	pF	1-4
R62	IOC 10	69	4	100	pF	1-4
R63	IOC 9	70				
R64	IOC 8	71	4	100	pF	1-4
R65	SCE/DESTN 11	72	8	150	pF	1-4
R66	SCE/DESTN 10	73	8	150	pF	1-4
R67	SCE/DESTN 9	74	8	125	pF	1-4
R68	SCE/DESTN 8	76	8	150	pF	1-4
R69	IOC 15	82	4	75	pF	1-4
R70	IOC 14	83				
R71	IOC 13	85				
R72	IOC 12	86	4	75	pF	1-4
R73	SCE/DESTN 15	87	8	150	pF	1-4
R74	SCE/DESTN 14	88				
R75	SCE/DESTN 13	89				
R76	SCE/DESTN 12	91	8	150	pF	1-4
R77	RD DATA 8	93	4	75	pF	1-4
R78	RD DATA 9	94				
R79	RD DATA 10	95	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R80	RD DATA 11	96	4	75	pF	1-4
R81	RD DATA 12	97				
R82	RD DATA 13	98				
R83	RD DATA 14	99				
R84	RD DATA 15	100	4	75	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R84
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R84
Capacitance measurement	NA	± 7.0 pF	R1
	NA	± 3.65 pF	R2
	NA	± 2.4 pF	R3
	NA	± 3.65 pF	R4,R5
	NA	± 3.45 pF	R6
	NA	± 4.6 pF	R7
	NA	± 3.65 pF	R8,R9
	NA	± 7.2 pF	R10
	NA	± 7.0 pF	R11
	NA	± 4.8 pF	R12

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 4.7 pF	R13
	NA	± 7.2 pF	R14
	NA	± 4.8 pF	R15
	NA	± 3.65 pF	R16
	NA	± 2.4 pF	R17
	NA	± 3.65 pF	R18-R23
	NA	± 4.9 pF	R24,R25
	NA	± 3.55 pF	R26
	NA	± 4.8 pF	R27
	NA	± 6.05 pF	R28
	NA	± 6.15 pF	R29
	NA	± 4.7 pF	R30
	NA	± 3.65 pF	R31,R32
	NA	± 3.55 pF	R33,R34
	NA	± 3.65 pF	R35
	NA	± 7.4 pF	R36
	NA	± 4.8 pF	R37-R44
	NA	± 3.55 pF	R45-R59
	NA	± 4.8 pF	R60
	NA	± 3.55 pF	R61
	NA	± 4.8 pF	R62-R64
	NA	± 7.1 pF	R65
	NA	± 5.85 pF	R66,R67
	NA	± 7.1 pF	R68
	NA	± 3.55 pF	R69-R72
	NA	± 7.1 pF	R73-R76
	NA	± 3.55 pF	R77-R84

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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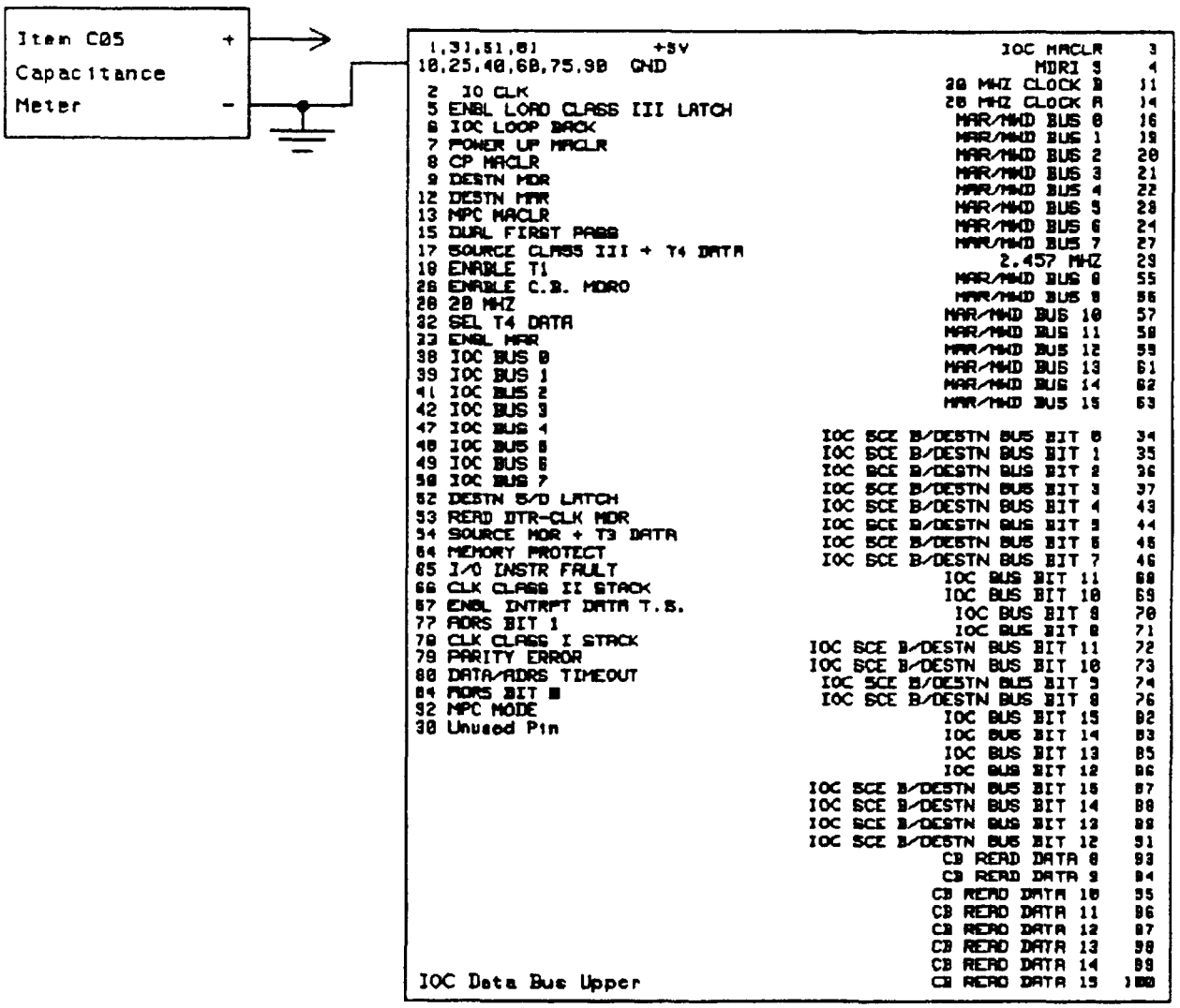


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.44 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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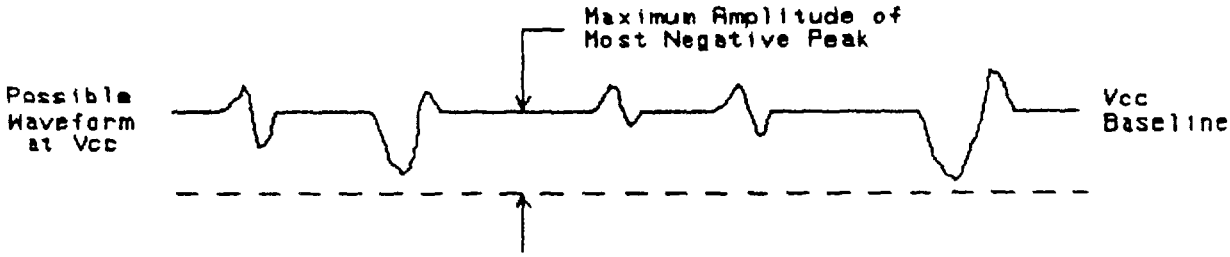
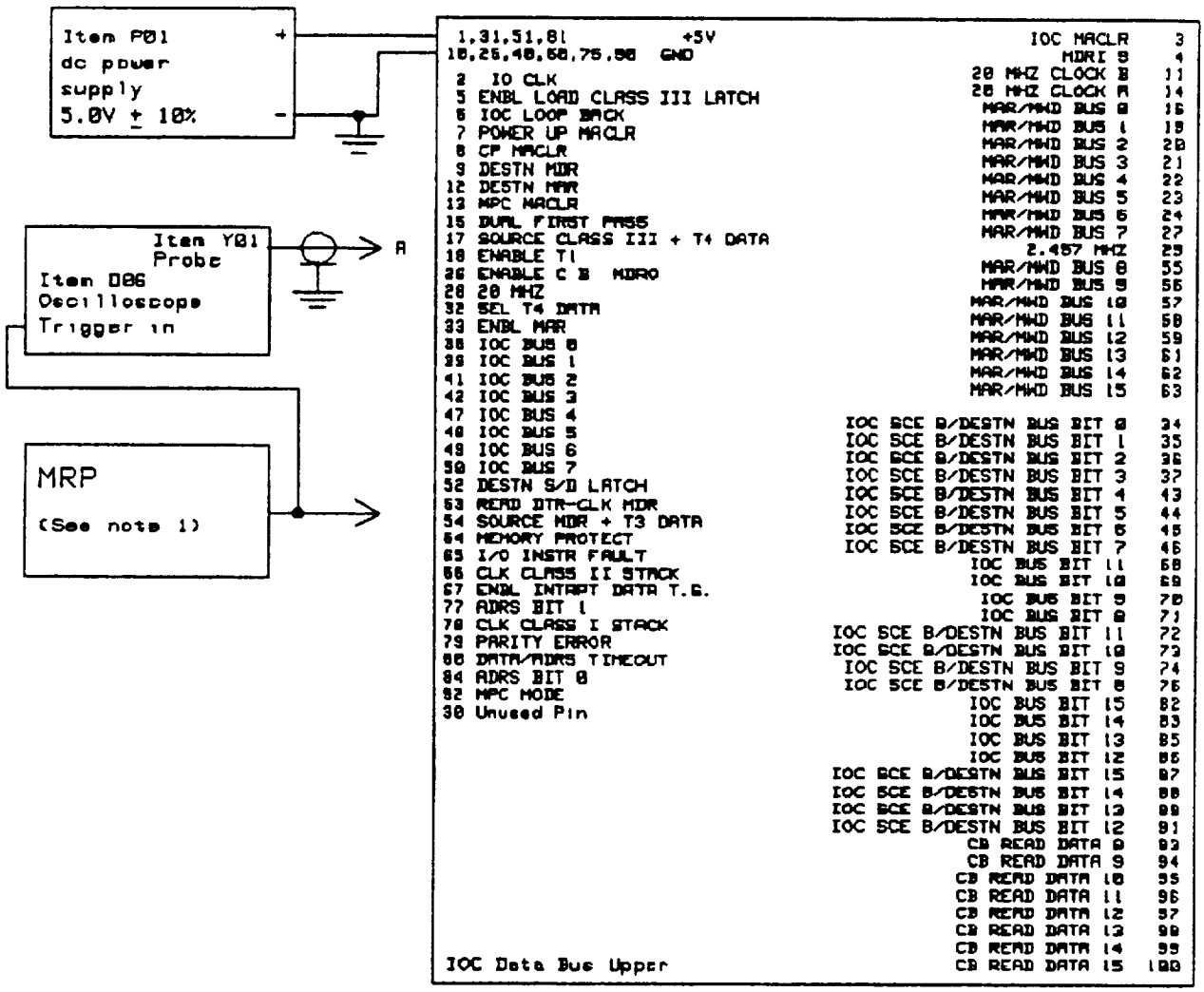


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-28787/535	Modules, Standard Electronic, IOC Data Bus Upper Module, Key Code EWC

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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